Why Did the Ranker Pick the Wrong Title? Explaining Recommendation Failures to Decide What to Fix

Counterfactual (TemGX, ICLR 2026) and causal (CATGX, KDD 2026) explanations over members' rating history

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[1] Lu et al., ICLR 2026. [2] Lu et al., KDD 2026.

One failed search, from the symptom to a fix worth testing

A search goes wrong
"a fantasy movie that isn't animated" → Aladdin is #1

Where does it fail?
Empirical study on a ranker I built

Which past events?
the few past ratings the model relied on (TemGX)

Which patterns?
recurring patterns that matter beyond timing (CATGX)

Which fix to test?
combine the evidence; check a fix before an A/B test

Demo and next steps
Live demo; where this can go

The investigation in five steps: it stays on the right of every slide as navigation

1 Failure | 2 Events | 3 Patterns | 4 Together | 5 Fix

Search = a member's request to the ranker, the model that orders titles. **Failed search** = the #1 title does not fit the request (Demo relevance labels).

One running case throughout: member 304 on public MovieLens data; the ranker and all Demo numbers are my own offline experiments.

Methods: [1] Lu et al., ICLR 2026; [2] Lu et al., KDD 2026. Data: MovieLens [5].

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What evidence does a model rely on?

My research: find which historical evidence a model's prediction depends on, check it by intervening on the same model, and turn it into a fix worth testing.

Q1 Which past events does one prediction depend on?

TemGX · Training-free Counterfactual Explanation for Temporal Graph Model Inference · ICLR 2026 [1]

Q2 Which recurring patterns matter beyond timing and activity?

CATGX · Causal-Aware Temporal Graph Explanation via Scalable Motif Sampling and Adjustment · KDD 2026 [2]

Q3 Does evidence-based explanation carry over to scientific images?

Context Determines Optimal Architecture in Materials Segmentation · ICLR 2026 AI4Mat workshop [13] · AFM image analysis, IMMI 2023 [12] · PGAD-X (ongoing)

Selected publications: [1,2,12,13]. Photograph supplied by Mingjian Lu. PGAD-X is ongoing research.

A member asks for a genre, with an explicit exclusion

“A fantasy movie that isn't animated”

A ranker adds three signals. Aladdin vs Harry Potter:

Text match (fits “fantasy”?): 0.99 vs 0.88

Popularity (how many watch it): 0.65 vs 0.47

History (recent ratings by this and similar members): 0.29 vs 0.23

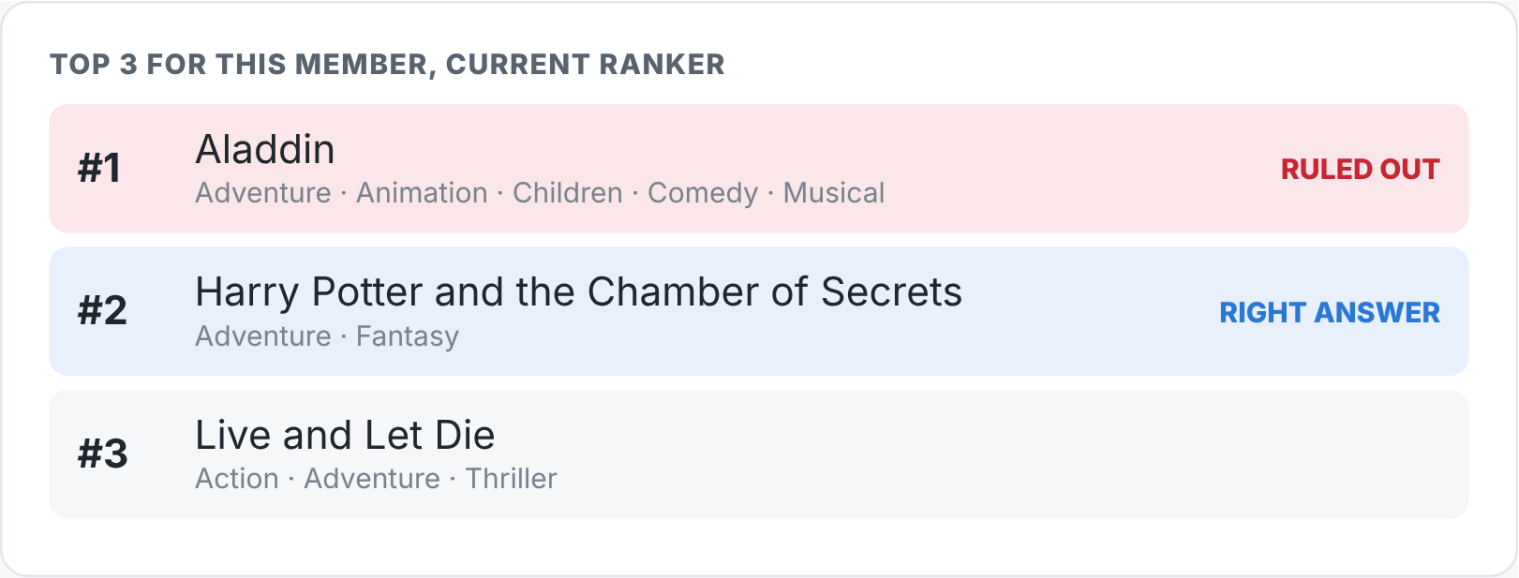
All three favour Aladdin, an animated film; none of them checks “isn't animated”



[3] Gomez-Uribe & Hunt, ACM TMIS, 2015. Signal values: the Demo ranker's standardized score parts for member 304 (MovieLens [5]).

The top result contradicts the member’s request

Member 304: “A fantasy movie that isn't animated”



#1

is animated, which the member ruled out; the right answer is #2

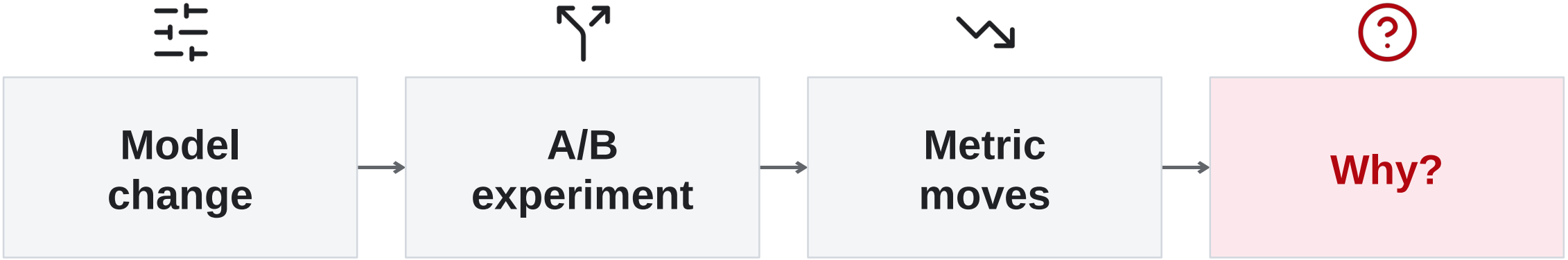
Right answer: a title that fits the request according to the Demo's relevance labels (here a live-action fantasy).

Why does Aladdin win, and what should the team change?

Dataset: MovieLens [5]. Demo, member 304.

An A/B test shows that a metric moved, not why Aladdin won

Model = the ranker: request + rating history in, ranked titles out. An A/B test compares two rankers on a metric, e.g. how often members play a top result.

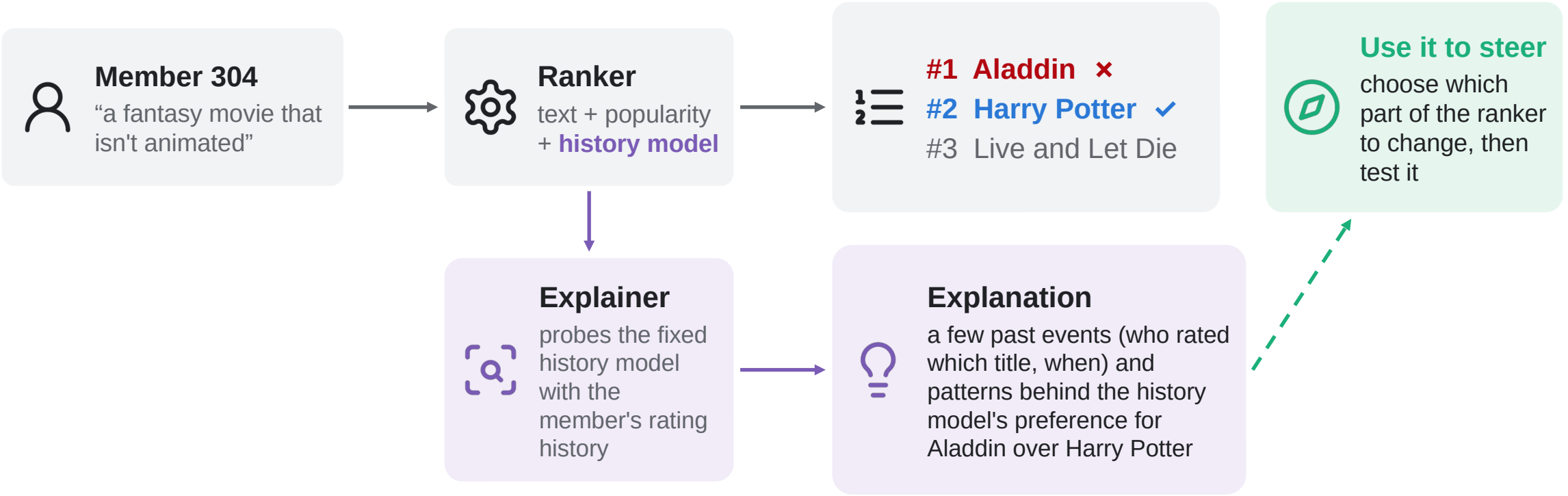


Which requests fail? Which past events made Aladdin win?

An A/B test reports one aggregate number over all members; it cannot say which of member 304's past ratings pushed Aladdin up.

[3] Gomez-Uribe & Hunt, 2015. [4] Kohavi et al., 2009.

Idea: explain the wrong ranking with the evidence it depends on



Discriminative: removing the explanation must change the fixed history model's preference between the wrong #1 and the right answer. The full ranking changes only when the team edits the ranker (step 5).

Illustration of the approach. Ranker and data: Demo on MovieLens [5].

A five-step investigation, from a failed search to a fix worth testing

1 Find the failure

Empirical study: which requests fail, how often, and which ranker signal caused it

In: logged searches
Out: failing requests

2 Which events?

TemGX: a compact set of past ratings whose removal changes the history model's preference

In: one failed search
Out: k events

3 Which patterns?

CATGX: recurring patterns that still matter when compared only with equally recent, equally active ones

In: rating history
Out: patterns that matter

4 Put together

Combine the events, the patterns and how much each ranker signal contributed

In: steps 1–3
Out: a diagnosis

5 Decide what to test

Reuse fixes that worked on similar, already-diagnosed failures; check on new requests

In: diagnosis
Out: a fix to A/B test

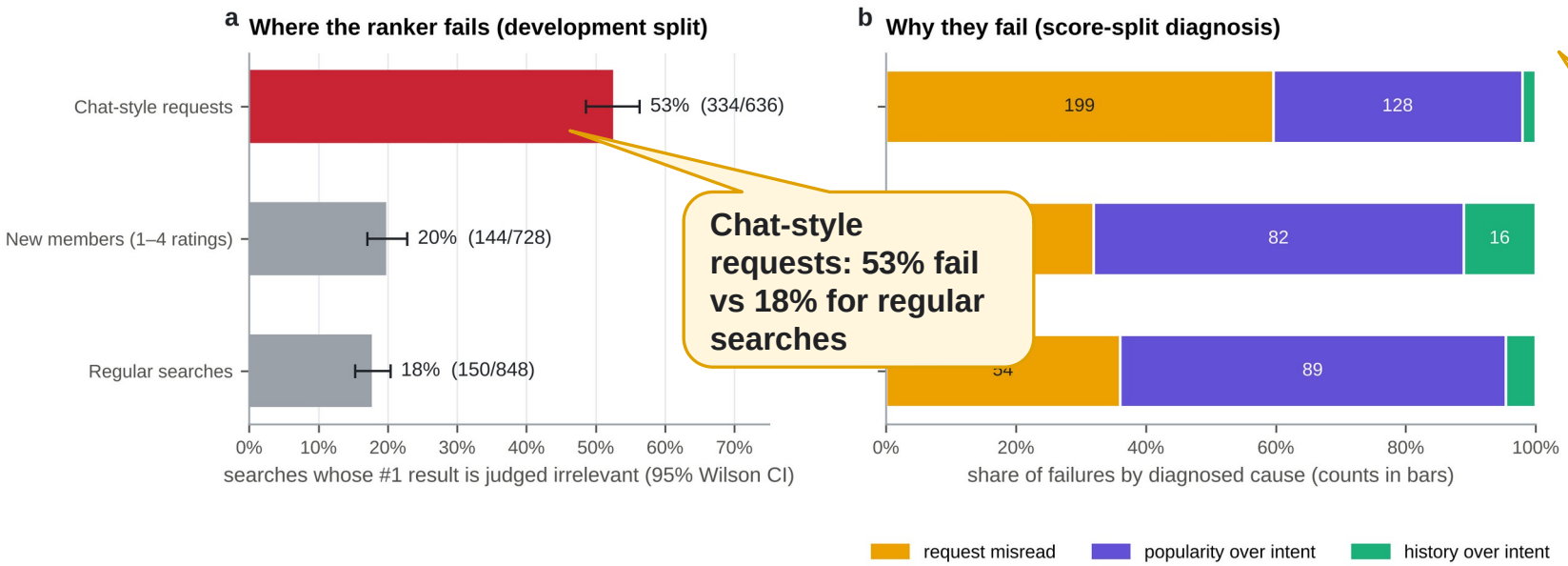
Each step can run as an agent: an event agent (step 2), a causal-analysis agent (3) and an explanation agent (4–5).

I built a Demo that runs all five steps on a failed search; I'll show it after the methods.

Methods: [1,2]. Demo on MovieLens [5].

Empirical study: chat-style requests fail three times as often

Setup: on public MovieLens I built a ranker (text match + popularity + a temporal graph model of rating history, the “history model”) and ran 2,212 searches. Failure: the #1 result is judged not relevant to the request (Demo relevance labels). Cause rule, fixed in advance: if text match favours the wrong title → request misread; otherwise whichever of popularity or history gives it the larger lead (a diagnostic rule, not a causal test).



Chat-style requests: 53% fail vs 18% for regular searches

Most failures: the request is misread or popularity overrides intent

So what: we know where it fails, but not which past events made the wrong title win

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Dataset: MovieLens [5]. Experiment: Demo, development split. Chat n = 636; regular n = 848; Wilson 95% intervals.

An important-looking event can be the wrong evidence



Member 304 · “a fantasy movie that isn't animated”

The model prefers **Aladdin** over **Harry Potter: 73%**

The 3 ratings with the highest attention weight:

- user 533 rated Aladdin 4.5★ · Apr 4, 2016
- user 138 rated Harry Potter 3★ · Jan 27, 2016
- member 304 rated The Usual Suspects 5★ · Apr 8, 2016

Remove all three → **74%**: the model still prefers Aladdin

Event = one rating: (user, title, stars, time). High attention weight is not the same as the evidence the decision depends on.

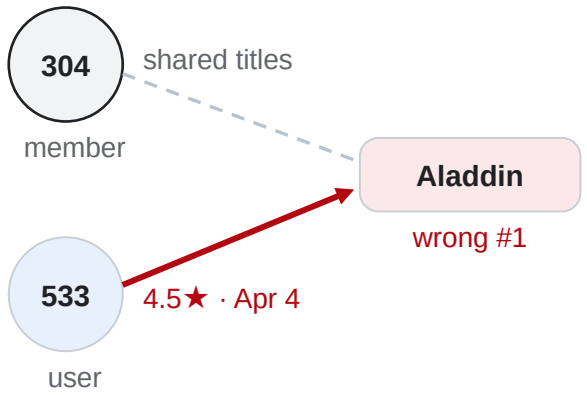
What evidence would explain why Aladdin wins?

1 Failure
2 Events TemGX
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[9] Jain & Wallace, 2019. [10] Wiegreffe & Pinter, 2019. Demo, member 304, three-rating removals.

Three questions, each answered by its own tool

Which events?

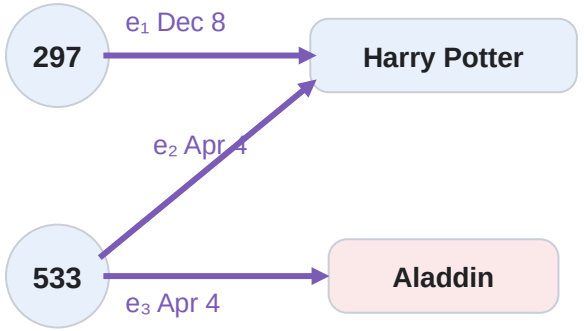


Single past events, e.g. one rating that made Aladdin look current

TemGX

finds a compact set of past events whose removal changes the history model's preference

Which pattern?



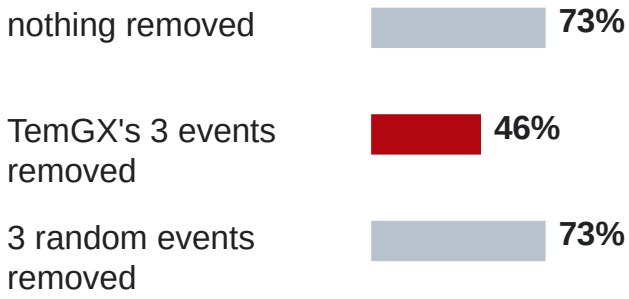
A recurring structure with an order ($e_1 < e_2 \leq e_3$): a Harry Potter fan's taste leads another member to Aladdin. Is it the structure, or just recent, active users?

CATGX

tests whether a pattern still matters after matching on timing and activity

Does it change the history model?

History model prefers Aladdin:



Below 50%, the history model ranks Harry Potter higher

Check

remove the evidence, re-run the same model, compare with random removals

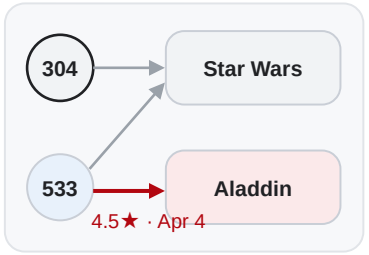


[1] Lu et al., ICLR 2026. [2] Lu et al., KDD 2026. Values: Demo on MovieLens [5], member 304.

Problem: which past events drive the history model?

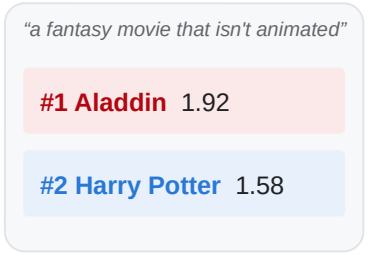
History graph G_t

Members and titles are nodes. Each edge is an event e = (user, title, stars, time τ), with τ before the search time t.



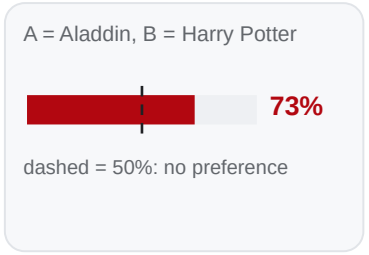
Ranker

score(i) = weighted sum of standardized text(q, i), popularity(i) and h(u, i; G_t); h is the history model, a 2-layer temporal graph attention network (TGAT).



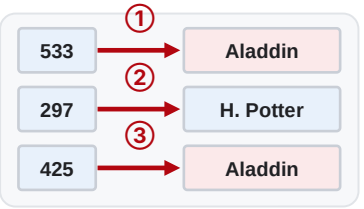
Target y

y = P(A above B) = σ((h(u, A) − h(u, B)) / T); A = wrong #1, B = right answer, σ = logistic, T = 0.65 (fitted on validation pairs).



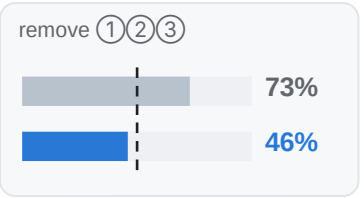
Explanation

A set of events $E_s \subseteq G_t$ with $|E_s| \le k$.



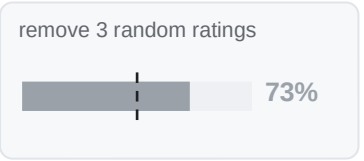
Counterfactual

Keep the trained model fixed; remove E_s; re-run. If $y(G_t - E_s) < 50\%$, the model now prefers B.



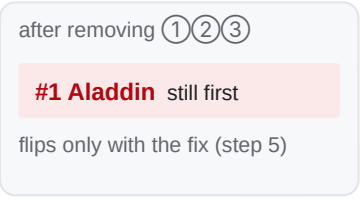
Control

Compare with random sets of k events from the same weeks.



Scope

y concerns the history model only; the full ranking also adds text and popularity. k counts explanatory nodes (here 3 users).



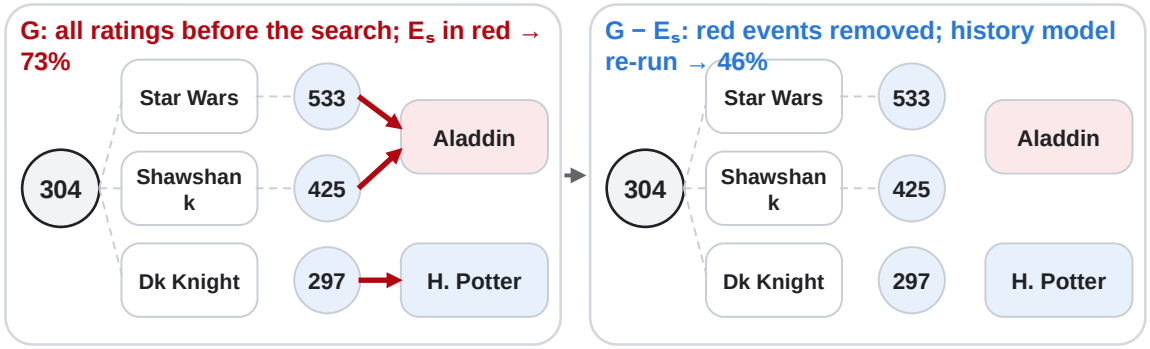
Formulation follows [1]. Mini figures: running values from the Demo on MovieLens [5], member 304.

Counterfactual explanation under a fixed model

Question

Would the same trained history model prefer differently if a small part of the history were absent?

Counterfactual explanation: a compact set of past events whose removal changes the fixed history model's preference.



Strict decision change

$$\hat{y}\theta(G - E_s) \neq \hat{y}\theta(G)$$

ŷ: does the history model prefer Aladdin over Harry Potter (y > 50%)? θ: its trained weights, fixed.
E_s: the removed events (red, left).

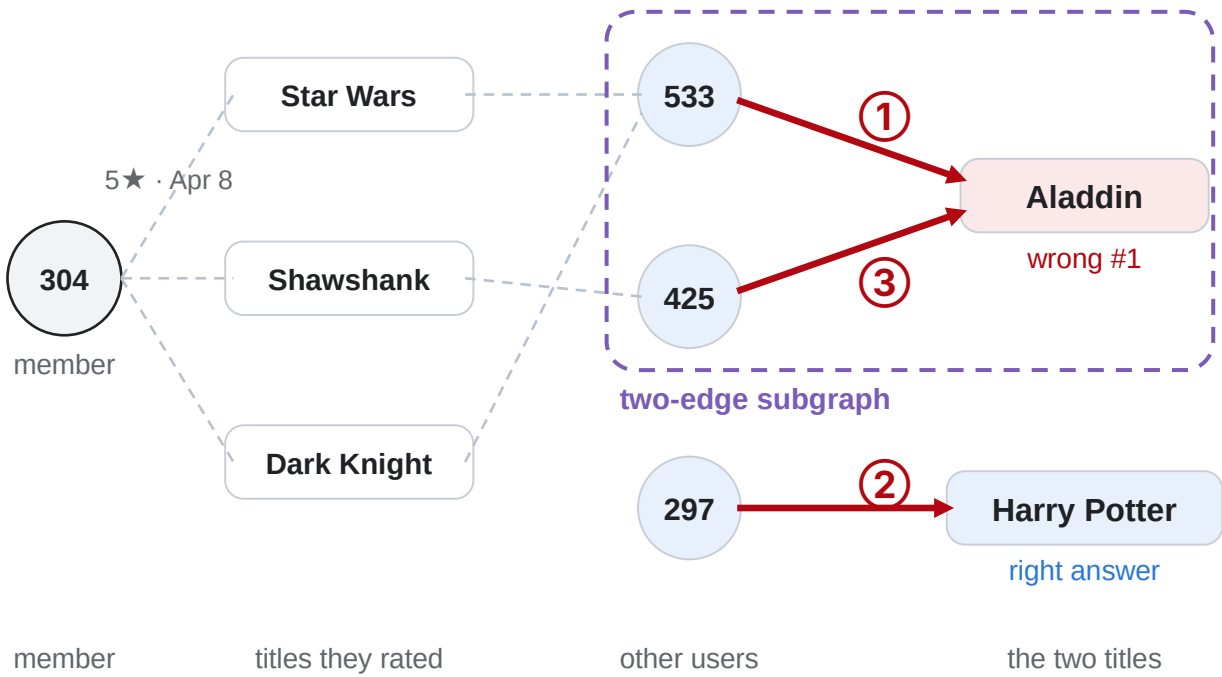
73% → 46%

Member 304: ŷ(G) = Aladdin at 73%; ŷ(G − E_s) = Harry Potter (46% for Aladdin)

1 Failure
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4 Together
5 Fix

[1] Lu et al., ICLR 2026. [6] Wachter et al., 2018. Example: Demo on MovieLens [5], member 304, k = 3.

TemGX finds a counterfactual explanation in terms of events



Selected events (k = 3)

① **User 533 rated Aladdin 4.5★, Apr 4, 2016 (4 days before)**

② **User 297 rated Harry Potter 3★, Dec 8, 2015**

③ **User 425 rated Aladdin 4.5★, Dec 27, 2015**

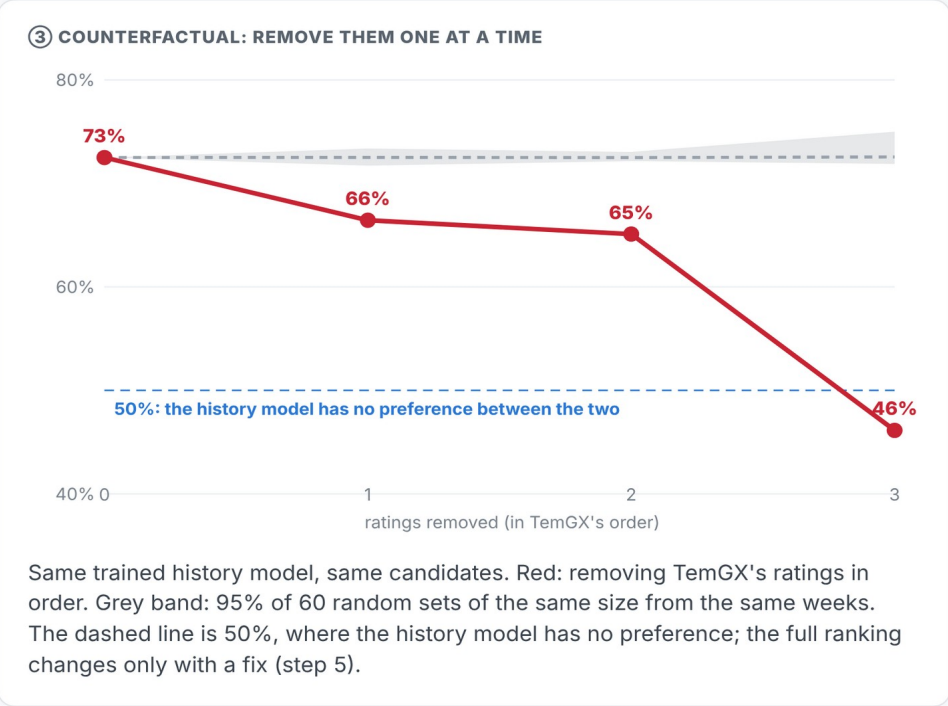
① + ③ act together. Two fans rating Aladdin months apart form a two-edge subgraph. Removed together: 73% → 48%; each alone: 66% and 72%.

Grey dashed: these users also rated titles member 304 rated, which ties them to the member.

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[1] Lu et al., ICLR 2026, §§3–4. Demo on MovieLens [5], member 304, k = 3. Removal values: same trained model, re-run with the events deleted.

Case study: removing 3 events flips the history model



Same trained history model, same candidates. Red: removing TemGX's ratings in order. Grey band: 95% of 60 random sets of the same size from the same weeks. The dashed line is 50%, where the history model has no preference; the full ranking changes only with a fix (step 5).

y-axis: the history model's probability of ranking Aladdin above Harry Potter (as on the problem slide); 50% = no preference. **x-axis:** TemGX's events removed one at a time; grey band: 60 random sets of the same size.

Full ranking, same request (top 3, total score)

Before removal	After removing TemGX's 3 events
#1 Aladdin 1.92	**#1 Aladdin** 1.87
#2 Harry Potter 1.58	**#2 Harry Potter** 1.58
#3 Live and Let Die 1.42	#3 Live and Let Die 1.42
History part, Aladdin − Harry Potter: **+0.05**	History part, Aladdin − Harry Potter: **−0.01**

History now favours Harry Potter, but text match and popularity still keep Aladdin first (lead 0.35 → 0.28).

The history model now prefers the right answer (73% → 46%; random sets: 73%). The full ranking flips only with the fix in step 5.

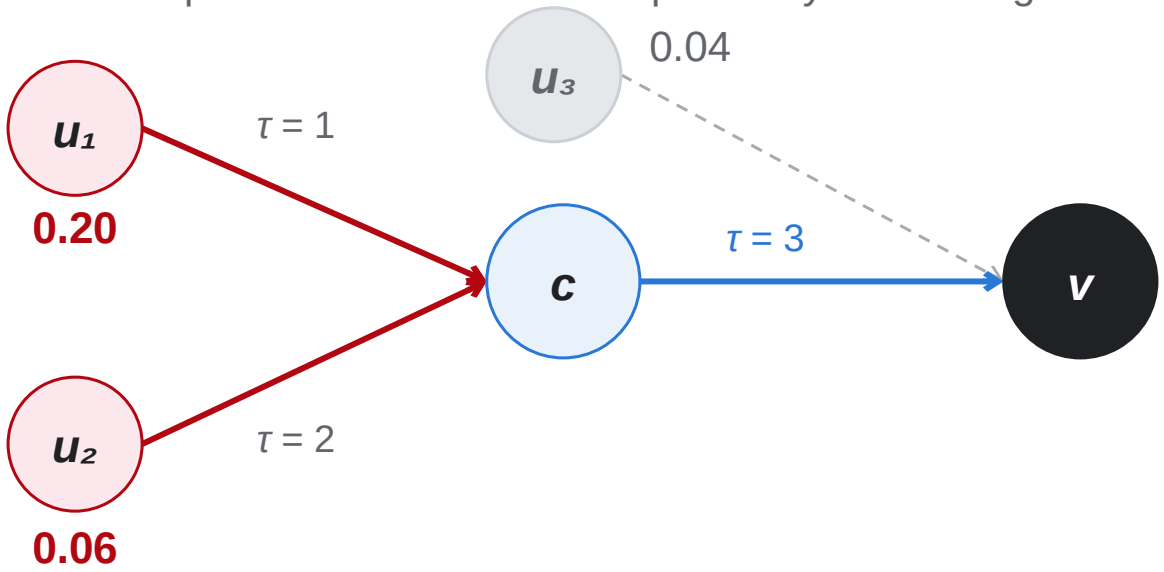
1 Failure
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[1] Lu et al., ICLR 2026. Demo on MovieLens [5]: member 304, k = 3; grey band and dots = 60 random sets of the same size.

TemGX picks explanatory nodes under a budget k

maximize $\Phi(V_s, v)$ **subject to $|V_s| \le k$**

Φ = total temporal influence k = explanatory-node budget



Illustrative temporal graph (teaching example). Times increase toward v.

Toy: V_s = {u_1, u_2} explanatory nodes (k = 2); connector c keeps a time-valid route to v; u_3 not selected.

The same optimization on member 304 (k = 3)

v (target): the history model's preference for Aladdin over Harry Potter, 73%

Candidates: 36 users and titles with Φ > 0, in a 2-hop window of 164 nodes and 200 rating events before the search

Greedy by Φ, each step verified by re-running the history model:

user 533 Φ 0.052 ✓ added

user 138 Φ 0.040 × removal did not lower the preference

…4 more rejected… user 297 Φ 0.007 ✓ user 425 Φ 0.006 ✓

V_s = {533, 297, 425}. Connectors Vc: none here, since each path is one rating straight into Aladdin or Harry Potter.

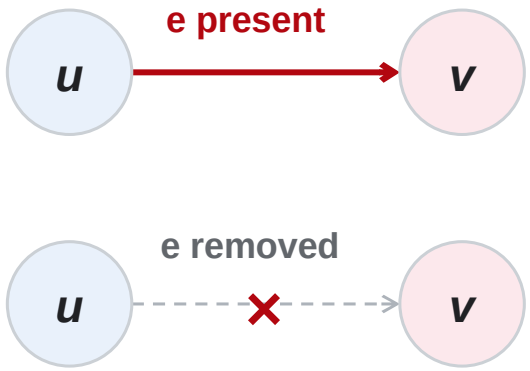
Budget: k = 3 nodes, giving 3 rating events.

[1] Lu et al., ICLR 2026, §3. Toy graph. k counts explanatory nodes; connectors preserve temporal paths.

Influence: remove one rating, then combine paths

$$p(e) = | \Pr(G,v) - \Pr(G \setminus \{e\},v) |$$

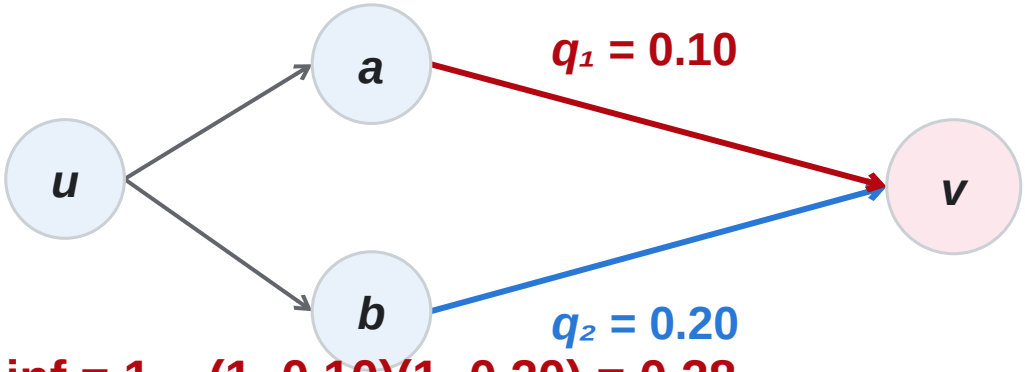
1 Remove one edge



$$p(e) = |0.80 - 0.70| = 0.10$$

2 Combine incoming paths

q = edge influence × upstream influence



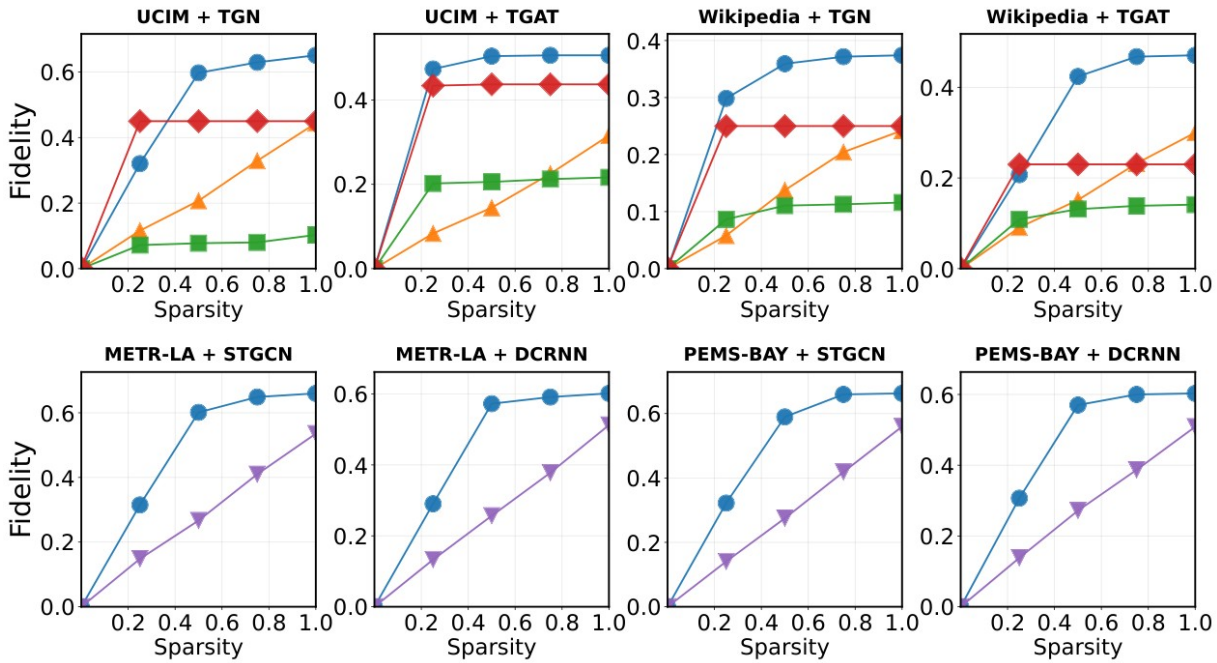
$$\text{inf} = 1 - (1-0.10)(1-0.20) = 0.28$$

Teaching values above. The same three steps on member 304, target v = the history model's preference for Aladdin (Pr = 0.725):
① **p(e), user 533 → Aladdin:** |0.725 − 0.664| = 0.061 (remove this rating, re-run the history model). ② **inf(533):** its time-valid paths combined = 0.063.
③ **Φ(533) =** 0.063 × resistance weight 0.83 × time decay 0.99 = 0.052, the highest of 36 candidates.

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[1] Lu et al., ICLR 2026, §3. Edge-removal sensitivity and ICM-style influence propagation. Illustrative values.

TemGX: fidelity across explanation budgets



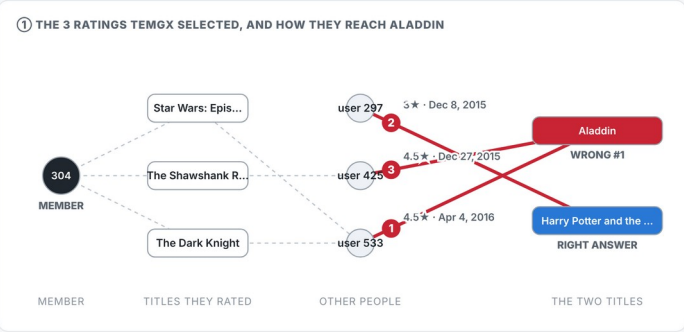
TemGX (blue) reaches the highest fidelity at most budgets, across 8 dataset–model pairs

Data: Wikipedia edits, UCIM messages, METR-LA / PEMS-BAY traffic sensors.
Protocol: matched backbones and explanation budgets, 5 independent seeds.

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[1] Lu et al., ICLR 2026, §5 and Fig. 4. Original fidelity panels; matched models and explanation budgets.

From selected ratings to a written diagnosis, with RAG



TemGX, k = 3: ① ② ③ are the red edges

Retrieved evidence, each rating checked by removal

①	User 533 rated Aladdin 4.5★ on Apr 4, 2016, **4 days before the search**; also rated **Star Wars** and **The Dark Knight**, which member 304 rated	remove only this: **−6.1 pts**
②	User 297 rated Harry Potter **3★** on Dec 8, 2015; it changes how Harry Potter is represented	remove only this: **−1.2 pts**
③	User 425 rated Aladdin 4.5★ on Dec 27, 2015; also rated **The Shawshank Redemption**, which member 304 rated	remove only this: **−0.6 pts**

All three removed: 73% → 46%; ① + ③: 73% → 48%; 3 random ratings: 73%.

RAG over verified diagnoses

1 Retrieve
TemGX + CATGX pick the evidence; HNSW finds the 3 closest of 343 verified diagnoses

→

2 Augment
One evidence sheet: the ratings, the pattern, the removal checks and the fixes that worked before

→

3 Generate
An LLM writes the diagnosis for the team from the sheet only, citing [1] [2] [3]

→

4 Check
Every number and citation must be on the sheet: 0 unsupported for member 304

Generated diagnosis, member 304: *"When Aladdin's ratings were removed, how sure the model was fell to 46%, so those ratings drove the result. … Try the fix "understand 'not' in the request", which worked for 9 of 10 of the most similar past failures, but trust it only moderately."*

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[1] Lu et al., ICLR 2026. RAG [14]; GraphRAG [15]. Demo on MovieLens [5], member 304, k = 3. Retrieval: HNSW over 343 verified DEV diagnoses. Generation: an LLM given only the evidence sheet, then an automatic number and citation check. Text shortened.

CATGX: patterns beyond timing and activity

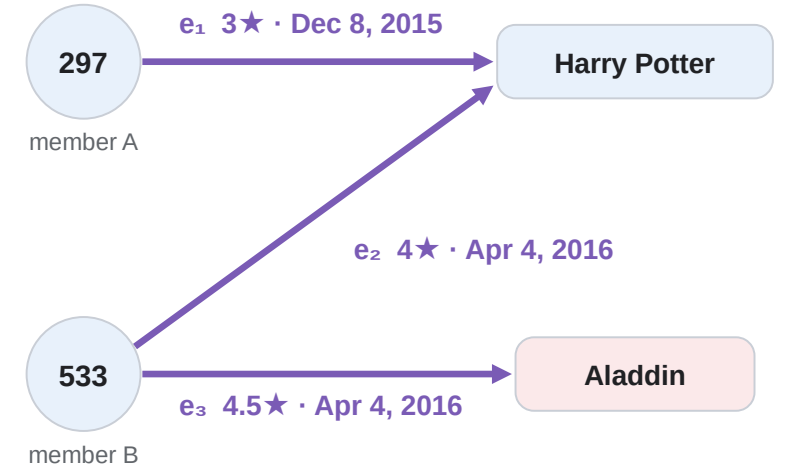
A temporal motif is a small labelled interaction pattern with a specified event order.

Pattern

Shared taste leads to the title: member B rates the same title as an earlier member A, then rates another title the same day.

Input: history, fixed model and motif library.
Output: a budgeted subgraph of motif occurrences.

Occurrence in member 304's neighbourhood



Order: $e_1 < e_2 \leq e_3$ (three edges, two members, two titles)

Does the pattern matter, or are the participants simply recent and active?

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[2] Lu et al., KDD 2026, §§2–3. [8] Paranjape et al., WSDM 2017. Case occurrence: Demo on MovieLens [5].

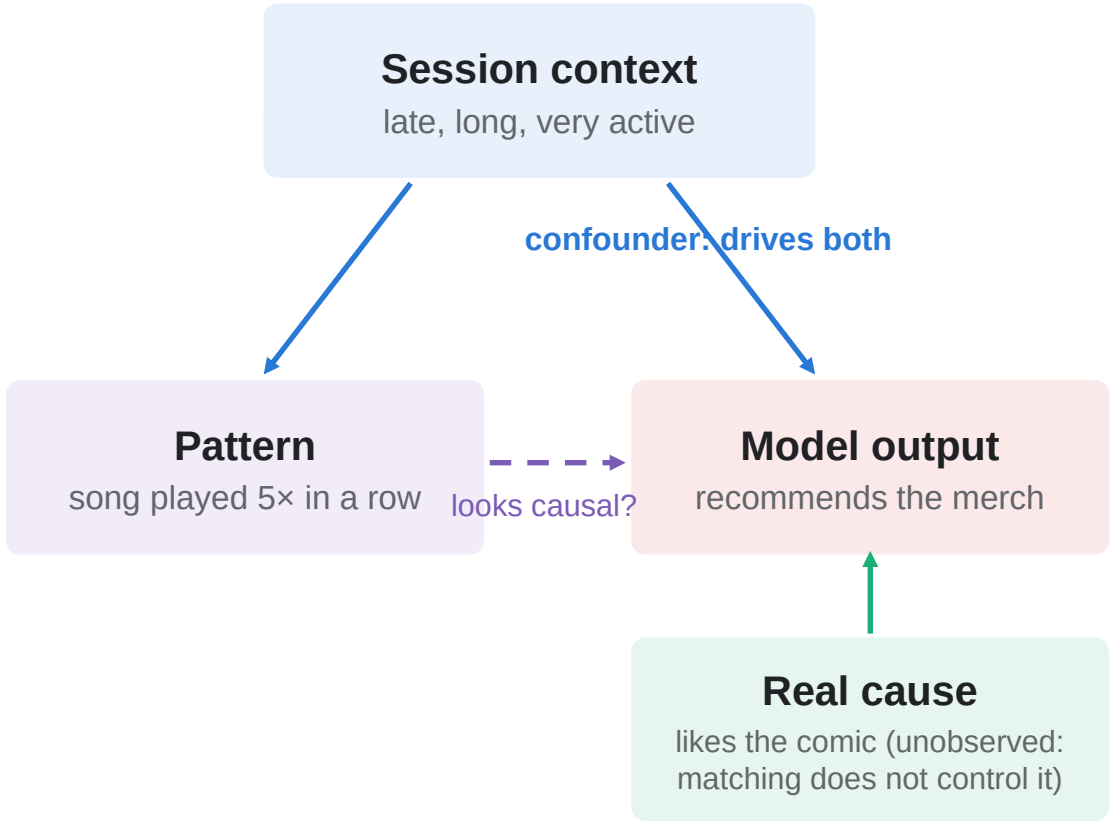
A pattern can look important partly because of its session

Illustrative example (not Demo data)

21:02	Rates a comic-based film 5★
21:30–22:10	Plays its theme song 5 times in a row
22:12	Same page shows a merchandise ad
22:15	Model recommends the merch title

The member is not interested in the merch: the recommendation is wrong.

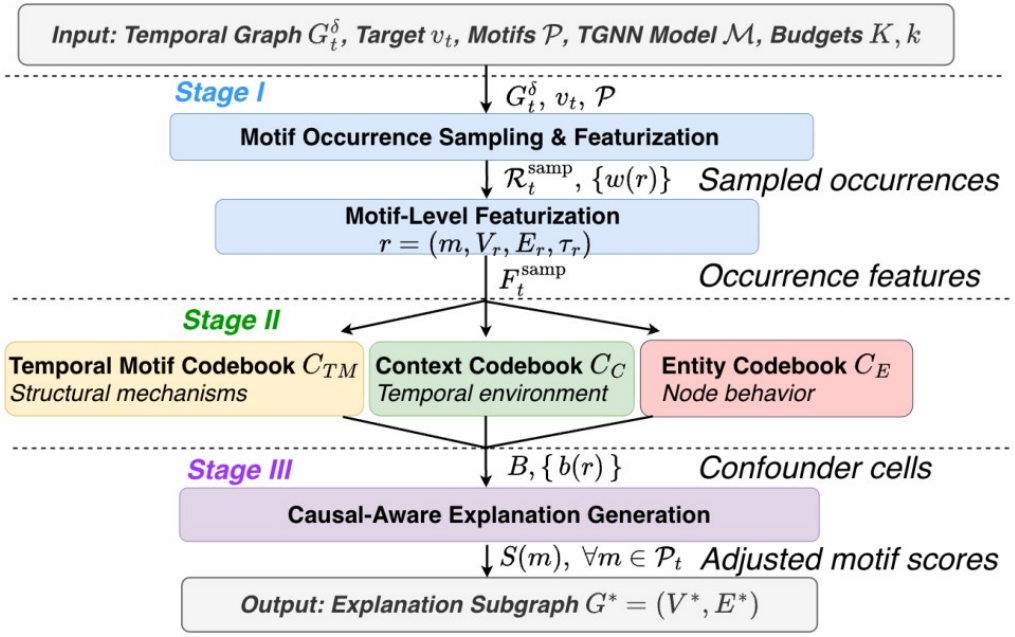
CATGX compares the pattern only with patterns from sessions of the same recency, length and activity. A lower adjusted score suggests that the measured context explains part of the pattern's apparent importance.



1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

[2] Lu et al., KDD 2026, §3 and §6. [7] Pearl, Statistics Surveys, 2009. Illustrative example; Demo values follow.

CATGX compares a pattern only within matched context cells



$$S(m) = \sum_b \omega(b)\,[\,\mu(m,b) - \mu_{base}(b)\,]$$

b: a cell of occurrences with the same recency, duration and activity bins (42 cells in member 304's window)

m = shared-taste motif
μ(m,b): mean effect of removing one of its occurrences in b

−

Baseline
μbase(b): same, over all patterns in b

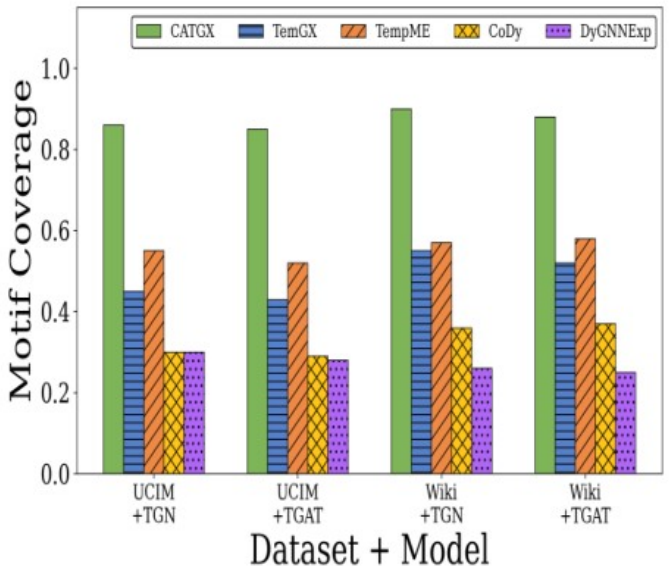
ω(b): share of sampled occurrences in b; sum over the 16 cells where m occurs

Member 304, shared taste: each removal costs 3.0 points of the history model's preference vs 0.9 for an equally recent, equally active pattern in the same cells (0.7 before matching). S(m) = 0.039 is this gap on the model's score scale.

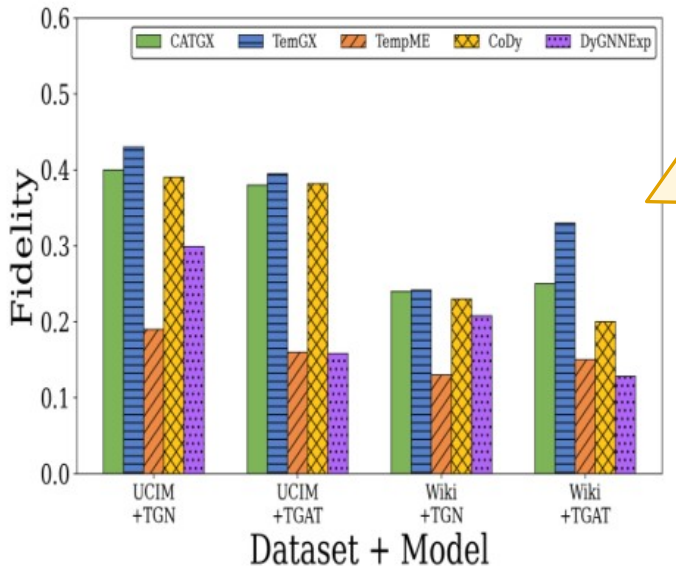
1 Failure
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3 Patterns CATGX
4 Together
5 Fix

[2] Lu et al., KDD 2026, Fig. 3 and §6, Algorithm 2. [7] Pearl, 2009, adjustment background.

CATGX: motif coverage and fidelity



(a) Overall Motif Coverage



(b) Overall Fidelity Comparison

CATGX recovers far more complete motifs (coverage ≈ 0.85–0.9 vs ≤ 0.6), at fidelity close to the strongest baseline

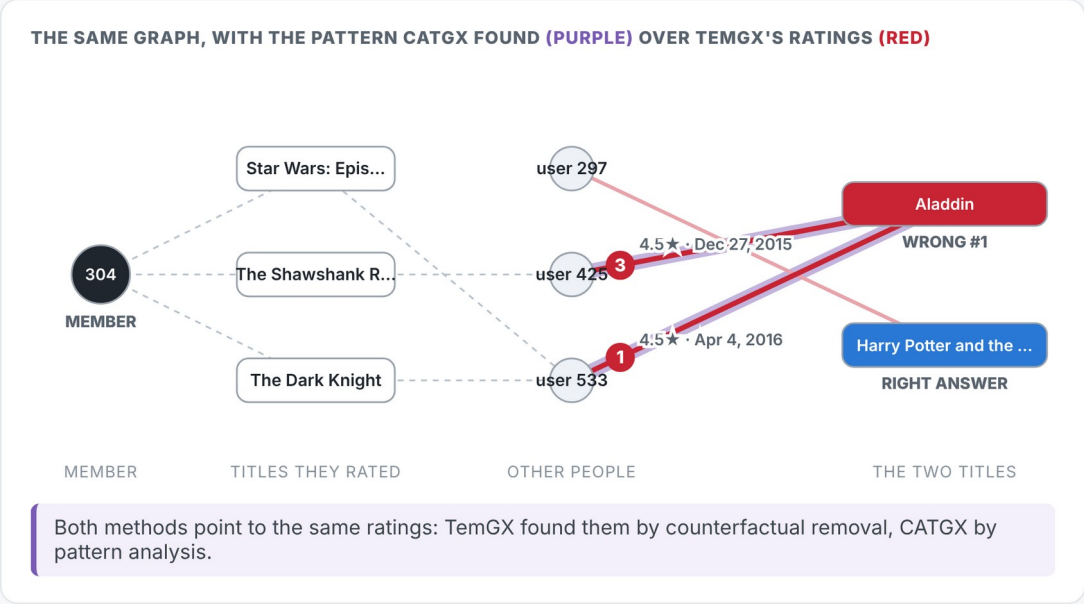
Data: Wikipedia edits / UCIM messages. Models: TGN / TGAT. Baselines: TemGX, TempME, CoDy, DyGNNExp.

200 test predictions, k = 50, aligned budgets. Coverage counts complete motifs; fidelity tests model-output behaviour.

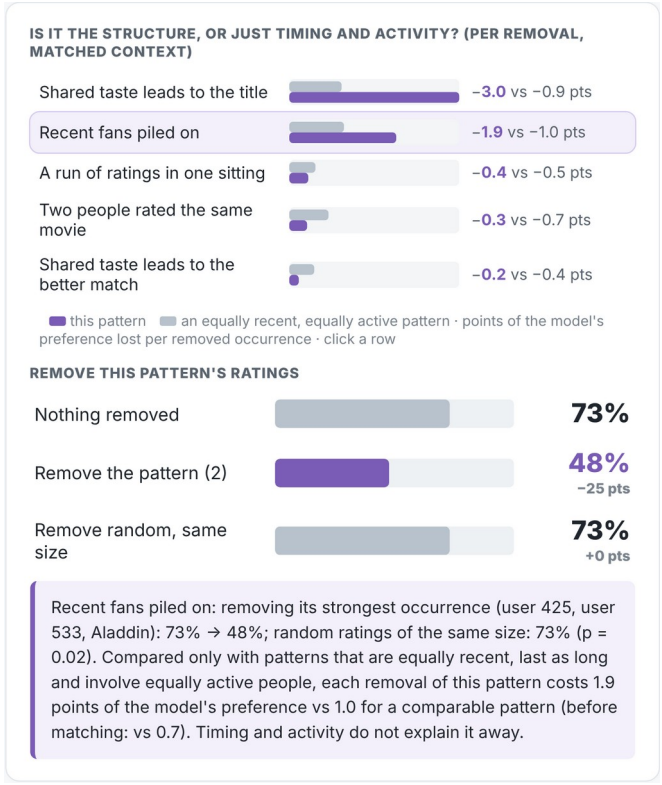
[2] Lu et al., KDD 2026, §7, Fig. 5(a–b). Original coverage and fidelity results.

1 Failure
2 Events TemGX
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4 Together
5 Fix

Two Aladdin patterns survive matching on timing and activity



Purple: the pattern CATGX found; **red:** TemGX's ratings. Both point to the same recent Aladdin fans.



73% → 48%

removing the two recent-fan ratings; random pair: 73% (p = 0.02)

Shared taste: 73% → 62%; random: 72%

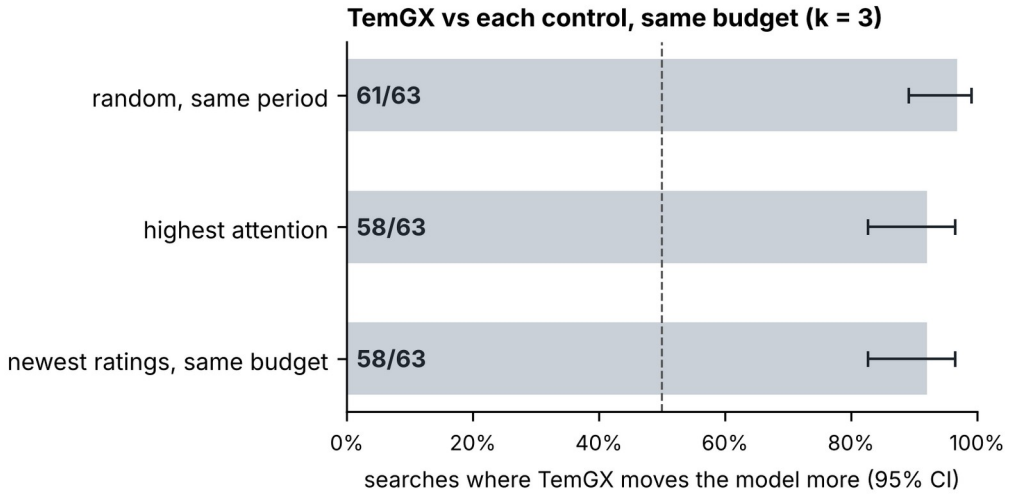
Per removal, matched on timing and activity: shared taste −3.0 vs −0.9 pts; recent fans −1.9 vs −1.0 pts

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

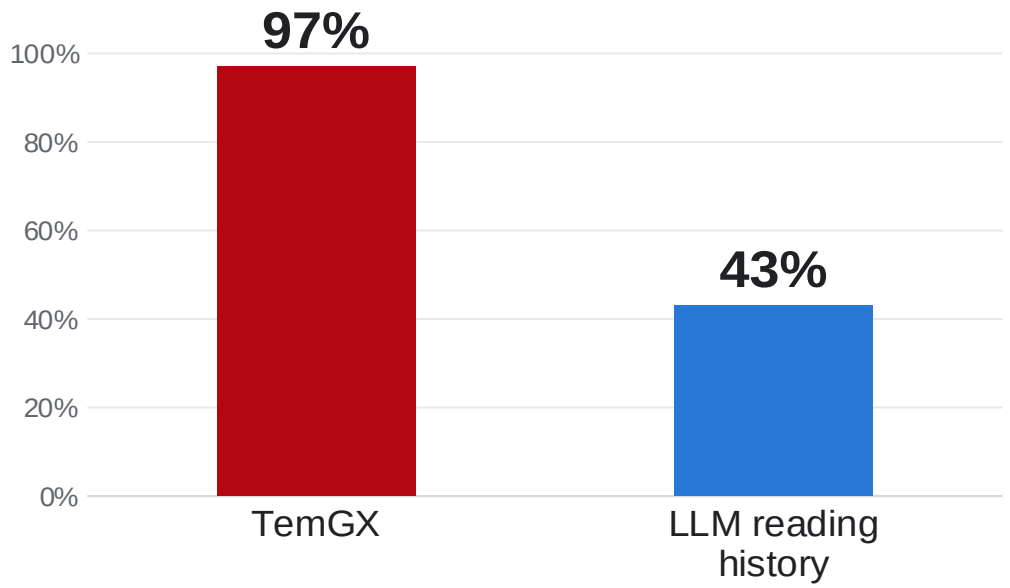
[2] Lu et al., KDD 2026. Demo on MovieLens [5], member 304.

Evidence must beat controls and recover a known effect

Deletion controls



Known planted mechanism



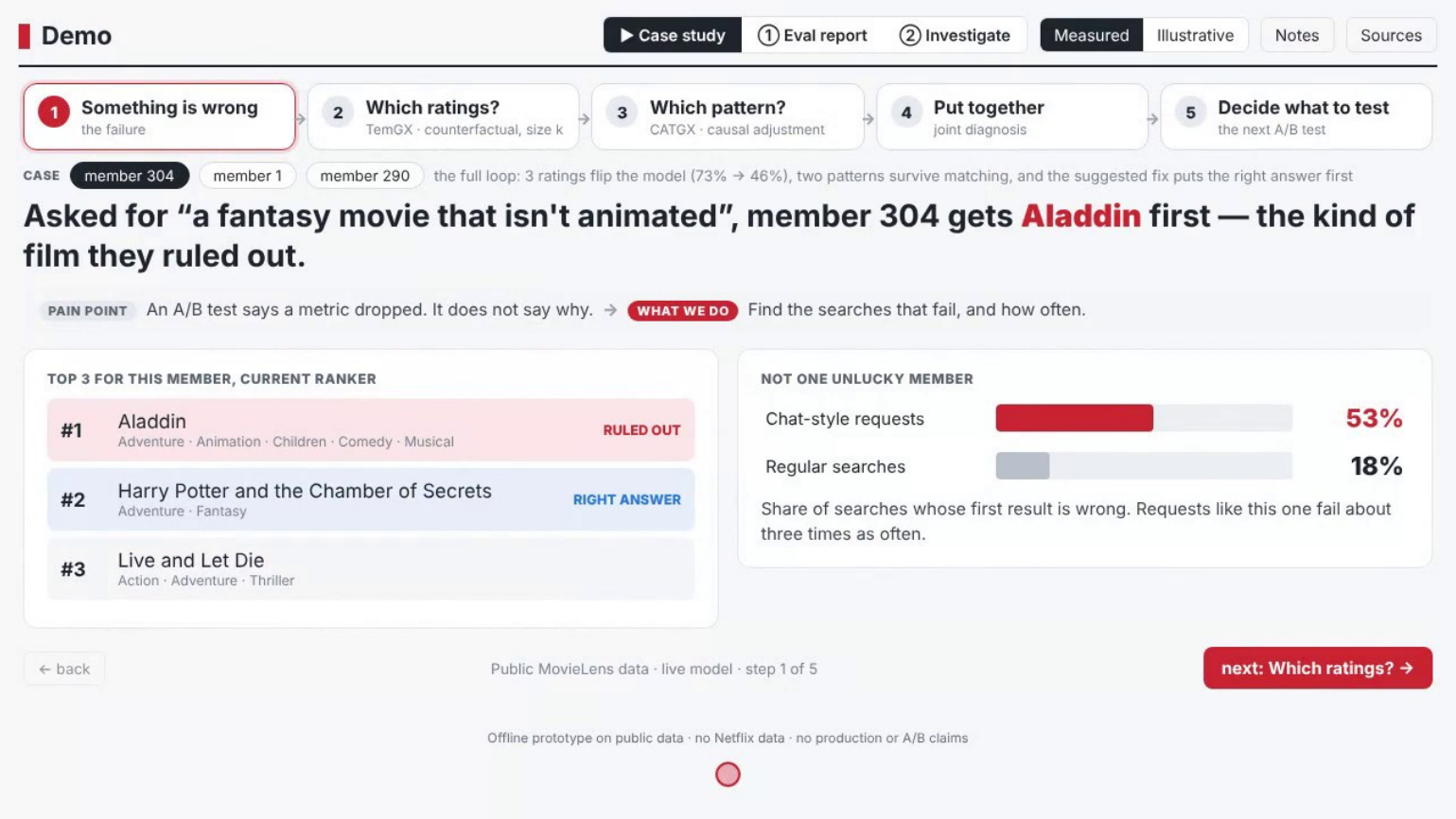
61 / 63 beat random deletion

97% vs 43% of the effect removed

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Demo on MovieLens [5]. Left: 63 failed searches for “a fantasy movie that isn't animated”, k = 3, Wilson 95% CIs. Right: 21 probes, median planted effect removed.

Aladdin comes first; the right answer is only second



PAIN POINT
An A/B test says a metric dropped. It does not say why.

WHAT WE DO
Find the searches that fail and how often.

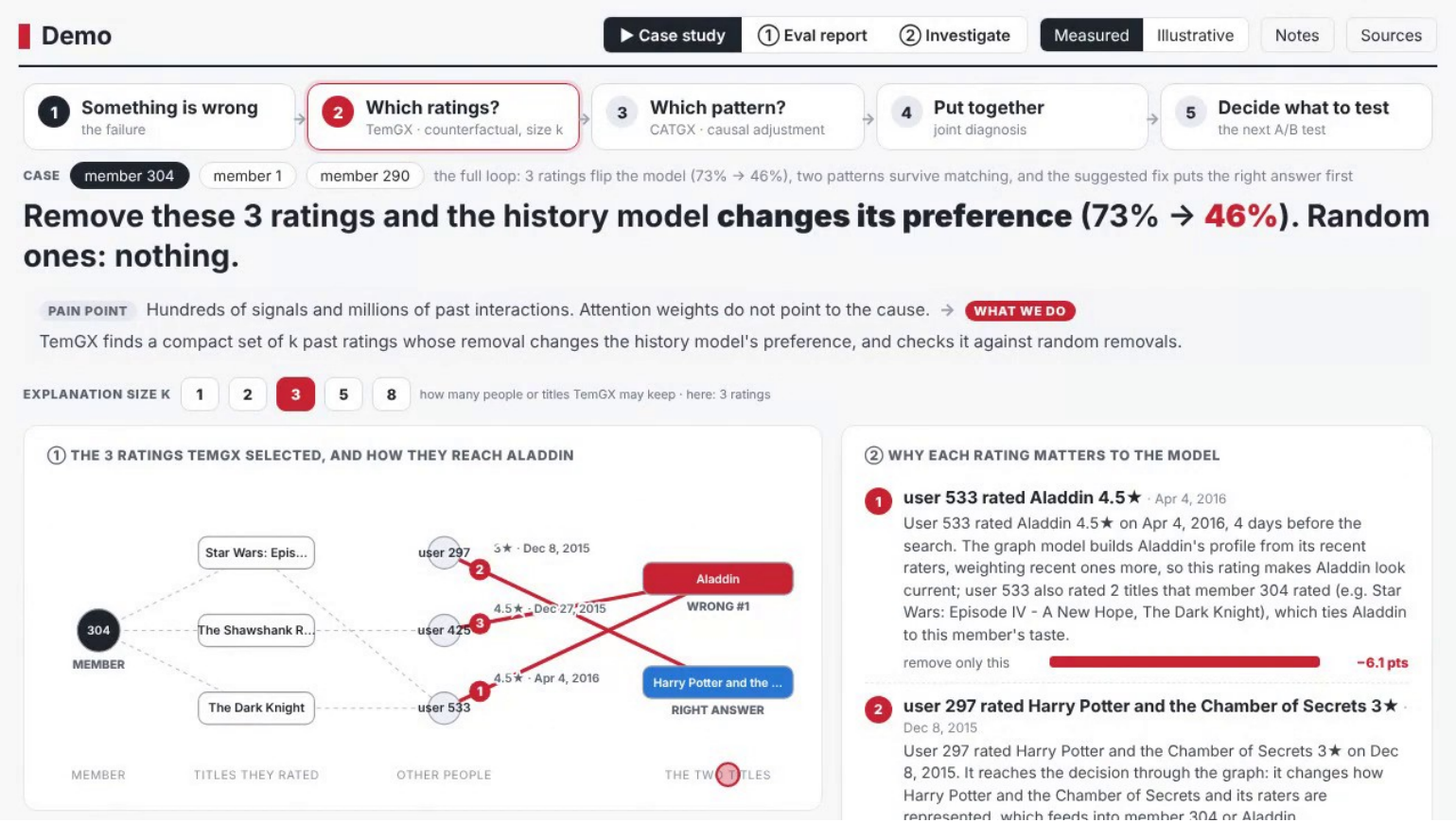
WHAT WE SEE
Member 304: Aladdin (animated) #1, Harry Potter and the Chamber of Secrets #2. Chat-style requests: 53% start with a wrong title, vs 18%.

Try it yourself
Code, models and data:
github.com/nicej1899/search-ranking-investigation-lab

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Dataset: MovieLens [5]. Demo.

TemGX finds and tests the ratings the model depends on



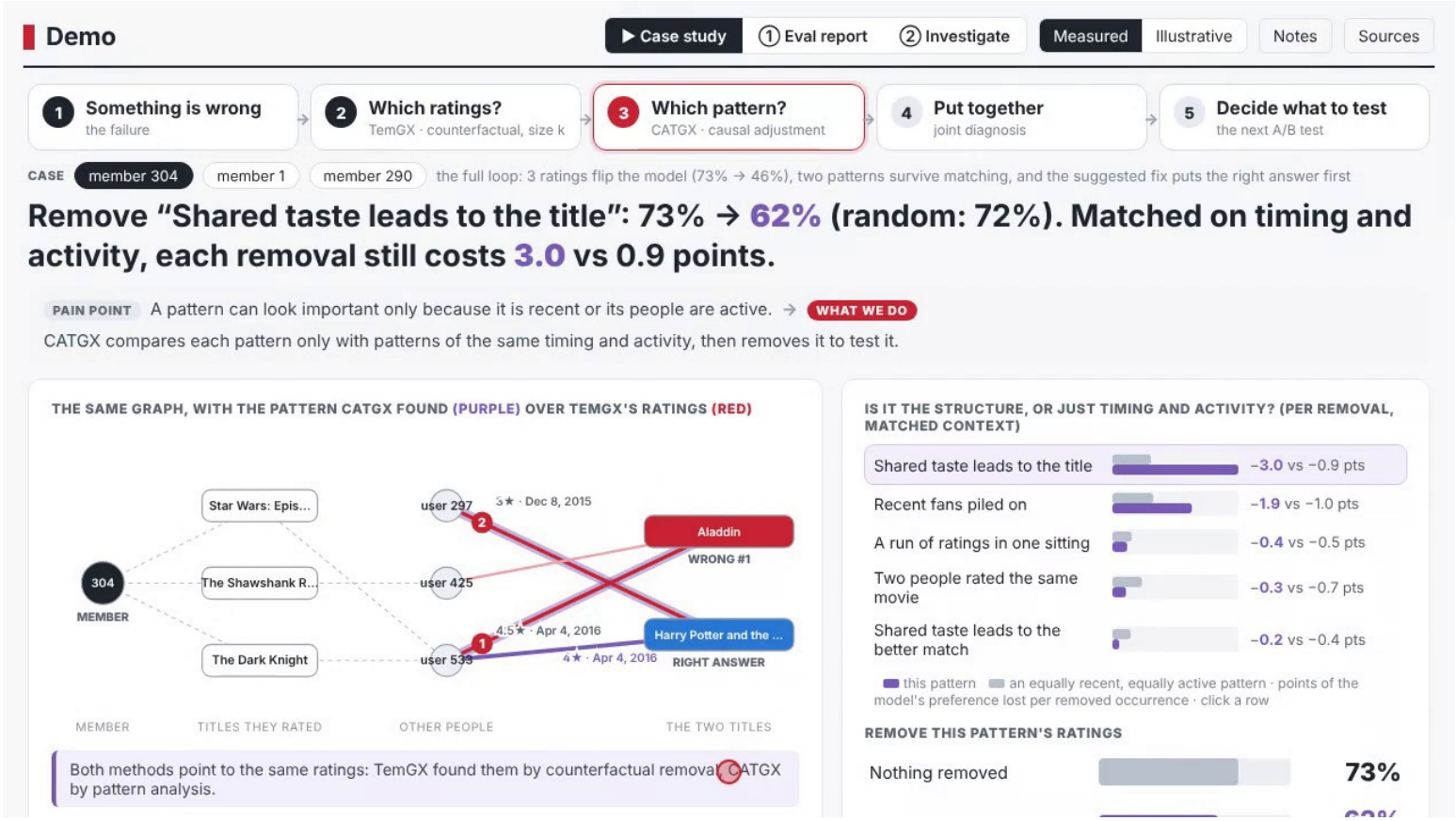
PAIN POINT
Hundreds of signals; attention weights do not point to the cause.

WHAT WE DO
Select a size-k explanation, then remove it and re-run the same model.

WHAT WE SEE
k = 3: 73% → 46%, the history model's preference flips; random sets: 73%; p = 0.016. Across 63 failed searches with this request: 61 beat random.

Dataset: MovieLens [5]. Demo.

CATGX checks whether a pattern survives matching



PAIN POINT
A pattern can look important only because it is recent or its people are active.

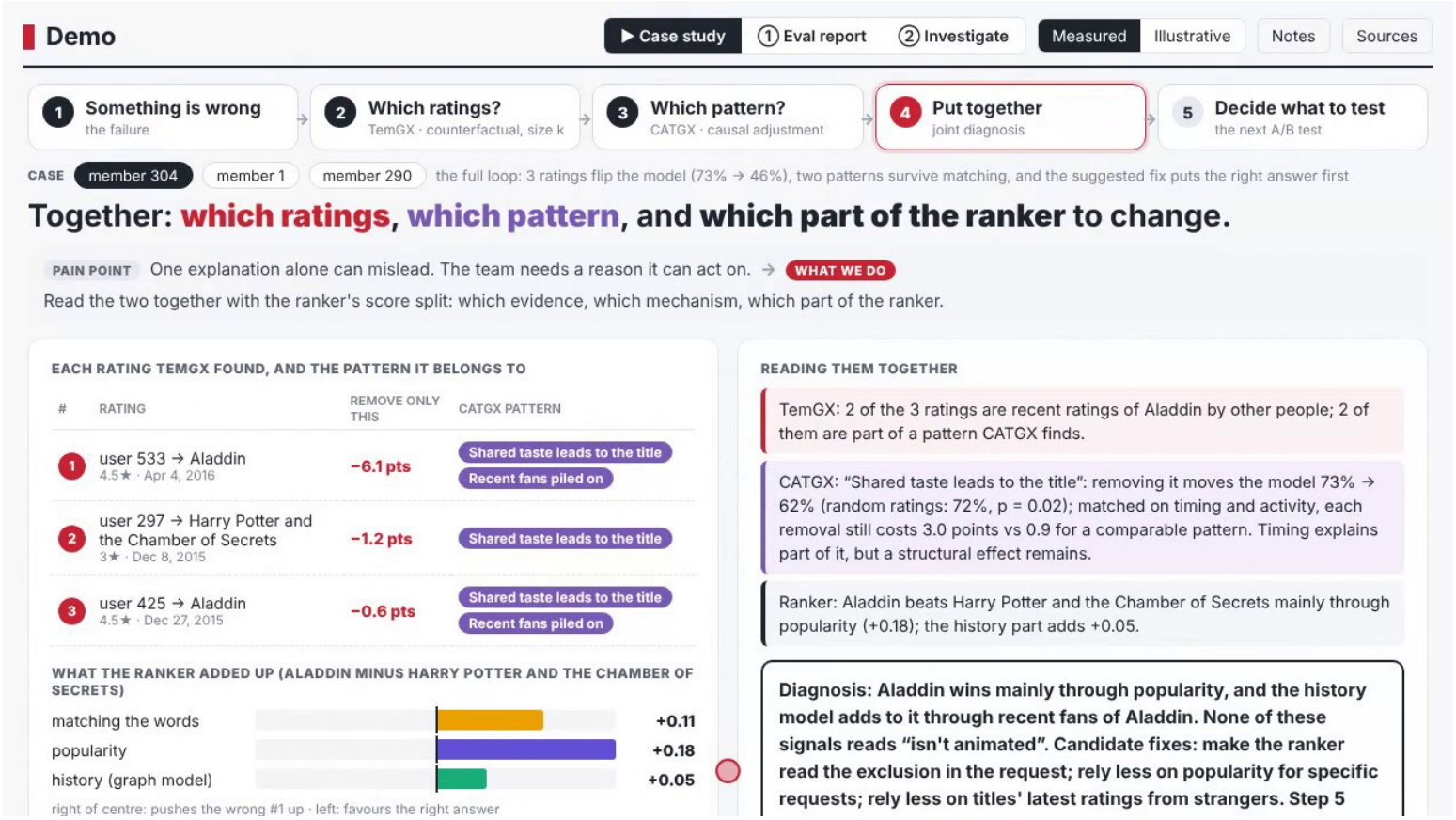
WHAT WE DO
Compare each pattern only with patterns of the same recency, duration and activity.

WHAT WE SEE
Removing shared taste: 73% → 62% (random 72%); the recent-fan pair: 73% → 48% (random 73%). Matched on timing and activity, each removal costs 3.0 vs 0.9 and 1.9 vs 1.0 points.

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Dataset: MovieLens [5]. Demo.

Which ratings, which pattern, which part of the ranker



PAIN POINT
One explanation alone can mislead. The team needs a reason it can act on.

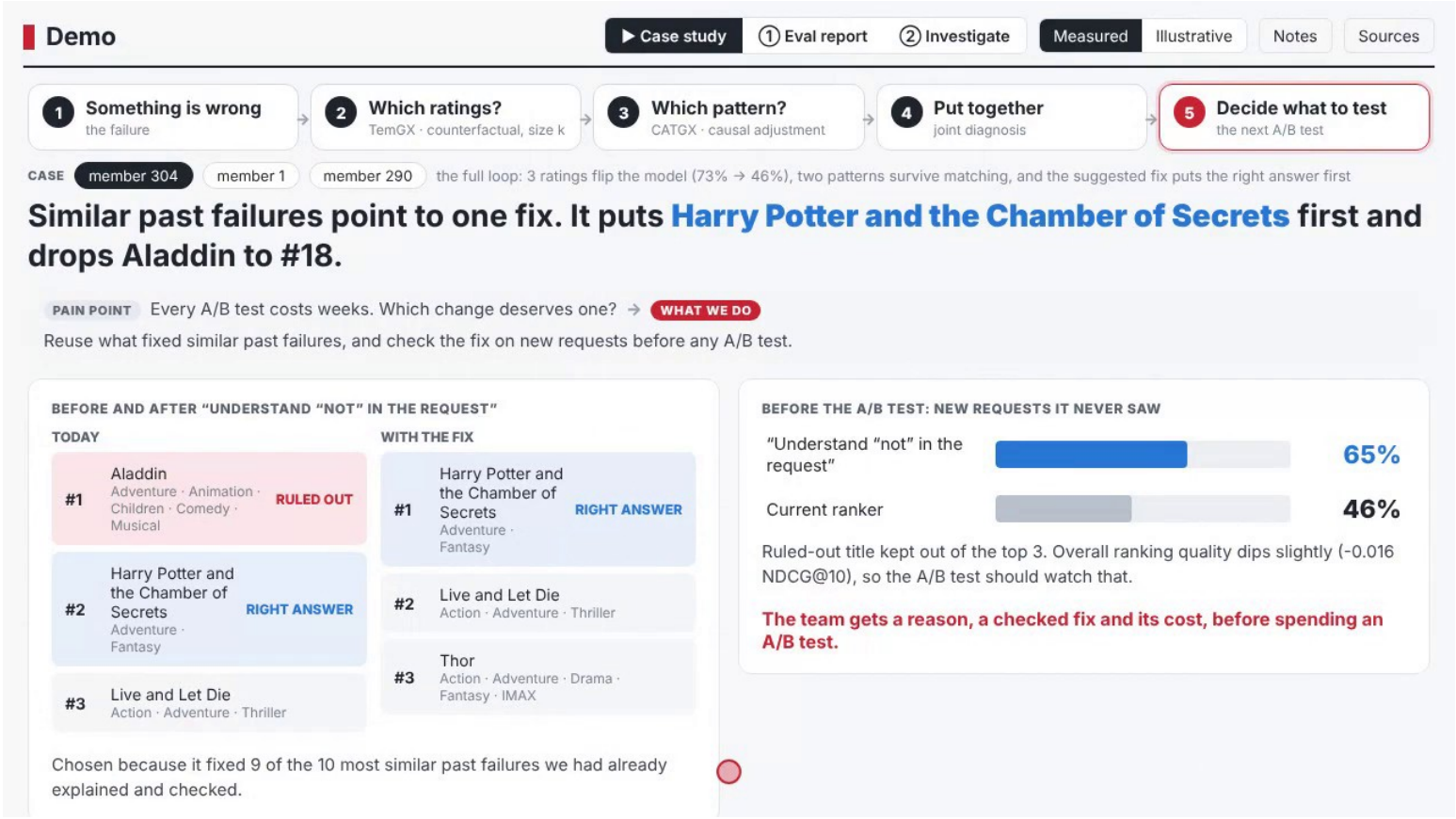
WHAT WE DO
Read TemGX, CATGX and the ranker's score split together.

WHAT WE SEE
Aladdin wins mainly through popularity (+0.18) and text match (+0.11); history adds +0.05 through its recent fans. Nothing reads "isn't animated".

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Dataset: MovieLens [5]. Demo.

A checked fix before spending an A/B test



PAIN POINT
Every A/B test costs weeks. Which change deserves one?

WHAT WE DO
Reuse what fixed similar verified failures; check the fix on new requests.

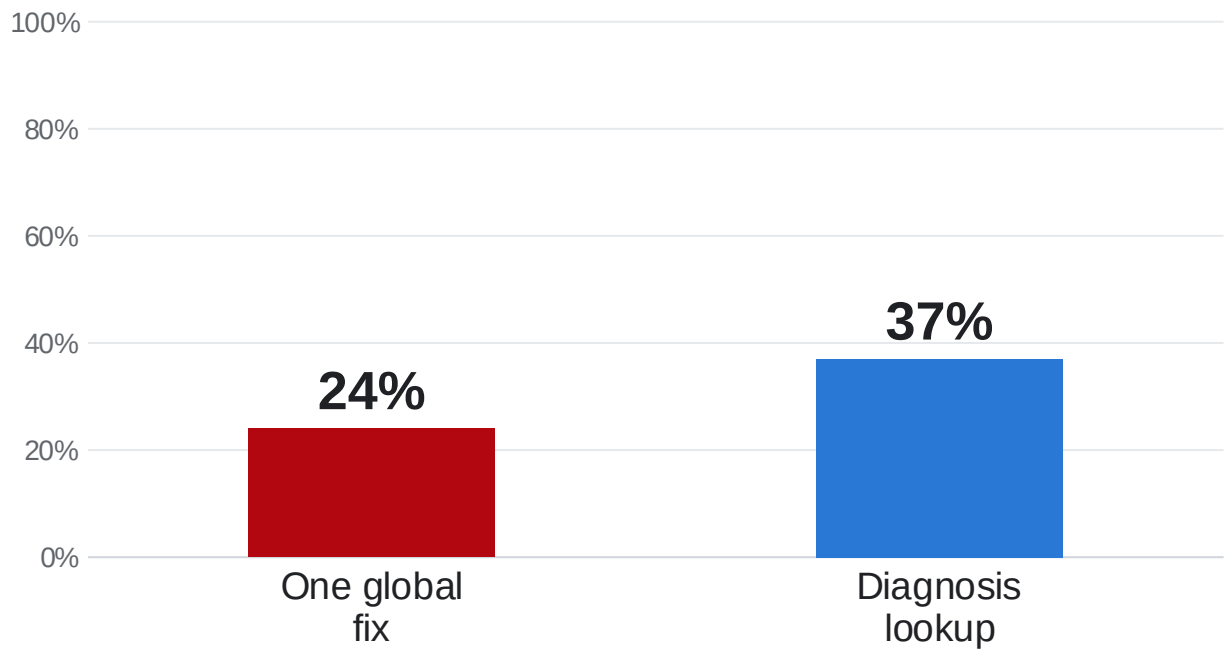
WHAT WE SEE
Recommended: understand "not" in the request (9 of 10 similar cases). Aladdin #1 → #18; Harry Potter → #1. Held-out: 65% vs 46% compliance; NDCG@10 −0.016.

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Dataset: MovieLens [5]. Demo.

Past verified failures help choose a repair

Share of 280 held-out failures fixed by the chosen repair



343

verified development diagnoses

280

new failures for evaluation

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

A repair candidate comes with comparable cases and measured outcomes

Experiment: Demo. 343 development diagnoses; 280 held-out failures. Paired gain +12.9 pp, bootstrap 95% CI [+8.6,+17.1]. Dataset: [5].

Where this could plug into the team's workflow

Model changes

1,726 regression checks from verified failures

Score a proposed change before an online test

Semantic IDs

22.9% of wrong-first results
vs 13.5% of candidates

Animation / Children proxy codes

Explain failure slices in item representations

Cold start

New-member NDCG@10 +0.014

95% CI [+0.010, +0.019]

How: for every candidate, drop its 10 most recent ratings by other members before the history score (explanations showed wrong winners propped up by strangers' recent ratings); 10 chosen on DEV, tested once on held-out FINAL.

Experiment: Demo. 1,726 checks / 576 requests. Semantic IDs: residual k-means proxy. Cold start: n = 528. Dataset: [5].

From explaining one failure to steering the ranker

Richer events

Multimodal and cross-platform actions joined into complex events, e.g. watches a trailer → plays the soundtrack → rates the film: one event the explainer can select.



Transfer across surfaces

Does evidence found on one surface, such as search, predict behaviour on another, such as the home page or games?



From explanation to intervention

Remove, down-weight or inject the evidence to steer the ranker, then test it offline before an A/B test, as in step 5.



Agentic diagnosis

An event agent, a causal-analysis agent and an explanation agent run the five steps on every new failure slice.



Goal: every failed search comes with a checked reason and a fix worth testing



[1] Lu et al., ICLR 2026. [2] Lu et al., KDD 2026. Directions are proposals for discussion. Code and Demo: github.com/nicej1899/search-ranking-investigation-lab

Backup guide: choose the question being asked

01 Definitions and causal meaning

02 Reading a temporal graph

03 Fixed-model intervention protocol

04 Budgets and the guarantee

05 Causal target versus adjusted score

06 Confounders, mediators and assumptions

07 A numerical adjustment example

08 Motif library and codebooks

09 HNSW and exact verification

10 Sampling weights and support

11 Deletion controls and budgets

12 Evaluation populations

13 Planted-effect metric

14 Attention and retrieval comparisons

15 Diagnosis retrieval and repair

16 Semantic IDs and cold start

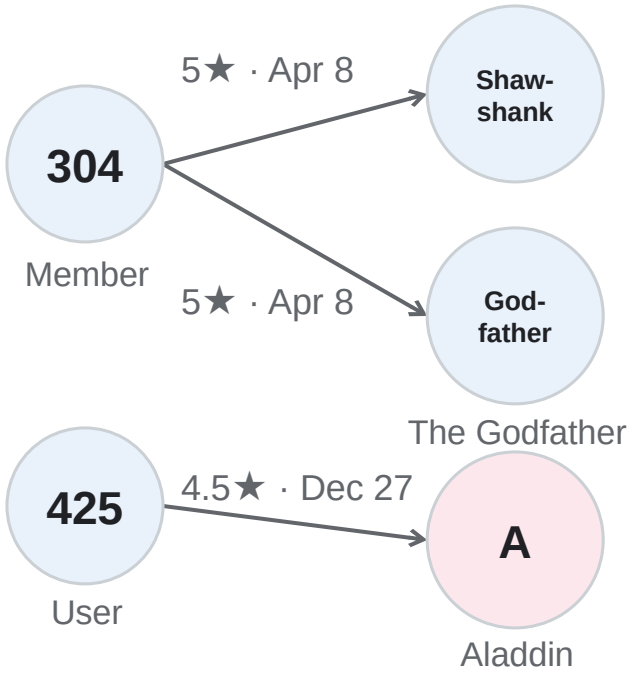
17 Runtime and deployment

18 My contribution and limitations

Published references are collected on the final page.

From timestamped history to a model prediction

Ghist: history observed before time _t_



$z = fE(Ghist)$ $y = fD(z)$

1 Encode

_fE_ Trained encoder

Summarizes temporal interactions

2 Represent

$z = [z_1, z_2, \ldots]$

Learned numerical representation

3 Decode

_fD_ Target score _y_

Example target: Aladdin versus Harry Potter and the Chamber of Secrets

History excerpt: ratings have values and timestamps. The full history goes into the same fixed model.

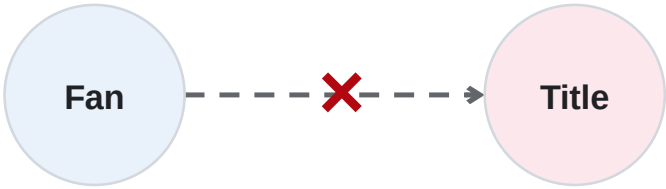
[1] Lu et al., ICLR 2026, §2. Example history: Demo on MovieLens [5], member 304.

One removal measures local sensitivity

Before



After removing one rating



Model probability: 0.80

Model probability: 0.70

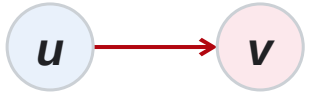
Influence = |0.80 − 0.70| = 0.10

[1] Lu et al., ICLR 2026, §3. [6] Wachter et al., 2018. Numerical values are a teaching example.

A node's influence is weighted by resistance and age

$$c(u,\tau) = \inf(u,v,\tau) \times g(\mathrm{TRD}(u,v,\tau)) \times \exp[-\lambda(t-\tau)]$$

u = historical node v = target τ = event time t = prediction time

	Influence	Resistance gate	Recency weight	Contribution
	 Temporal paths carry influence	 Higher resistance gives a smaller g	 Older evidence receives less weight	
Toy factors at one timestamp				
u_1	0.28	× 0.80	× 0.90	= 0.2016
u_2	0.28	× 0.40	× 0.50	= 0.0560

Average c over the time window, then sum over selected nodes to obtain $\Phi(V_s,v)$.

[1] Lu et al., ICLR 2026, §3. Temporal resistance and time decay. Toy factors, not experiment results.

Selection keeps explanatory nodes and their temporal connectors

1 Form candidates

Use *L*-hop temporal neighbours inside each sliding window.

2 Select and verify

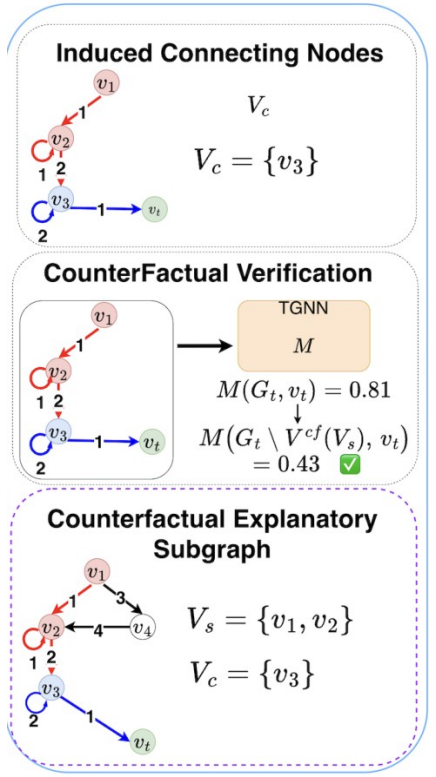
Rank by Φ, test candidate evidence with the fixed TGNN, and update the selected set.

3 Preserve temporal paths

Add connector nodes *Vc* and induce the explanatory subgraph.

4 Continue across windows

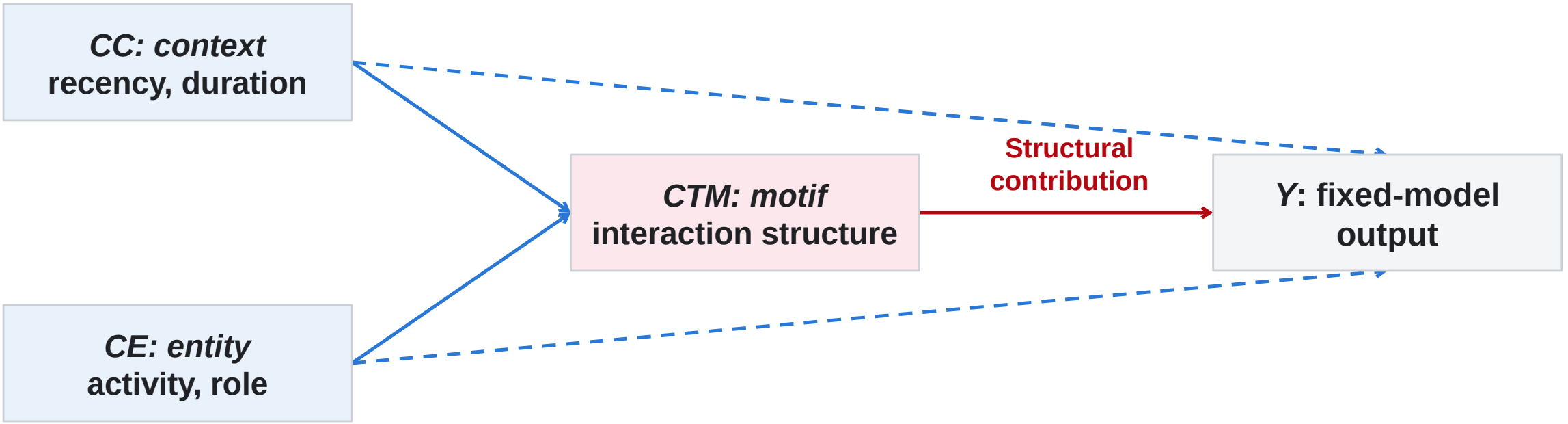
Update the explanation as historical interactions evolve.



The paper gives a (1−1/e) guarantee for its defined influence objective under the stated assumptions.

[1] Lu et al., ICLR 2026, §4 and Lemmas 1–3. Guarantee concerns the stated influence objective and assumptions.

Three codebooks separate structure, context and entity behaviour

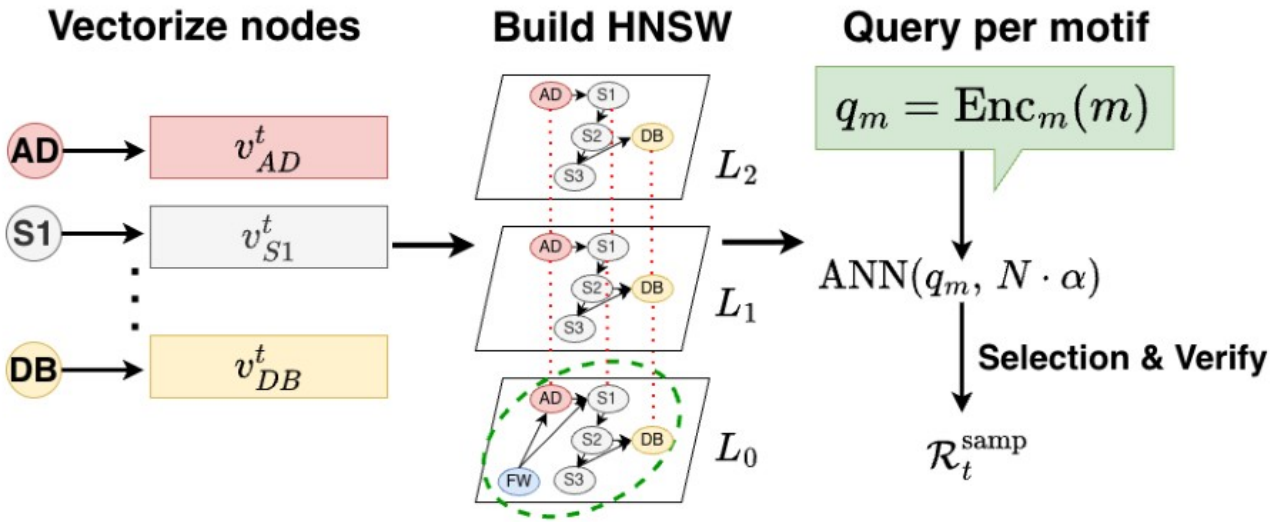


Movie example: compare rating patterns at similar times and activity levels

Target: dependence of the model output, conditional on the stated causal assumptions.

[2] Lu et al., KDD 2026, §3, Eq. (1), Fig. 2. [7] Pearl, 2009. Arrows specify the assumed model, not an identified effect on member behaviour.

Verified retrieval and sampling make motif selection practical



Verify candidates

Check node and edge labels and temporal order in the original graph.

Stratify and weight

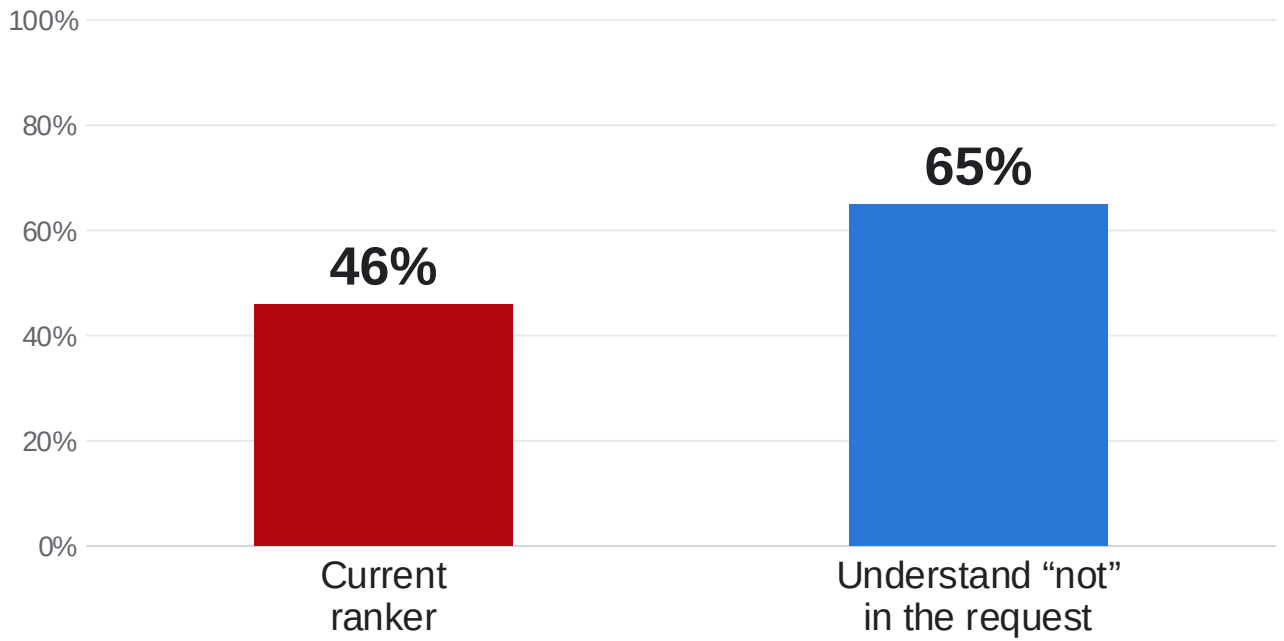
Group by recency, duration and activity. Use $w(r) = 1/\pi(r)$.

Occurrence selection: rank by $w(r)g(r;M) / |Er|$ and retain complete motifs under budget.

Full-population unbiasedness requires exact retrieval and positive support. ANN limits the covered population.

[2] Lu et al., KDD 2026, §§5–6. [11] Malkov & Yashunin, IEEE TPAMI, 2020. Exact-retrieval and positive-support assumptions govern unbiased totals.

A better constraint check can still carry a quality cost



65% vs 46%

excluded title kept out of the top three

−0.016

change in NDCG@10

Measure the repair’s benefit and its ranking-quality tradeoff

Experiment: Demo. Compliance: 576 held-out chat requests. Ranking quality: 468 requests, member bootstrap. Dataset: [5]. Evaluation background: [3,4].

Explanation, counterfactual and causal inference answer different questions

Explanation

Which observed inputs does this fixed predictor depend on?

Counterfactual

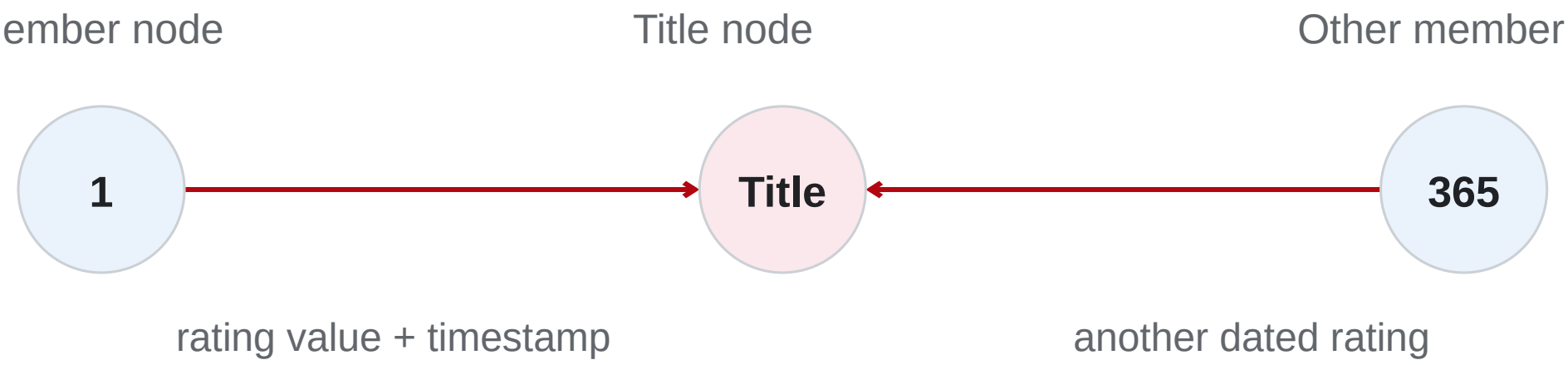
Would a permitted change to those inputs alter its decision?

Causal interpretation

Which dependence remains after adjustment under an explicit causal model?

[1,2,6,7] Published sources listed on the final References page.

A temporal graph records who interacted, with what, and when



Temporal neighbourhood + fixed prediction target + history cutoff

[1,5,8] Published sources listed on the final References page.

The intervention protocol holds the trained predictor fixed

Allowed change

Remove the selected historical events, keeping the target and model parameters fixed.

Required state handling

Recompute the affected representation from the permitted history. Stateful temporal models require consistent replay.

Leakage check

Use only events before the target time. Match the intervention and its controls.

[1,2] Published sources listed on the final References page.

The paper budget, Demo budget and approximation guarantee are distinct

TemGX formulation

k counts selected explanatory nodes; required connector nodes preserve temporal reachability.

Demo deletion test

Most controls remove three rating events. The CATGX case contains a two-rating occurrence.

Guarantee

The (1−1/e) bound applies to the defined influence objective under the paper’s assumptions. It is not a universal minimal-flip guarantee.

[1,2] Published sources listed on the final References page.

The causal estimand and the implemented adjusted score are different

Causal target under the assumed graph

$$P(Y \mid do(M = m)) = \sum_c P(Y \mid M = m, C = c)\, P(C = c)$$

Implemented CATGX score

$$S(m) = \sum_b \omega(b)\, [\mu(m,b) - \mu base(b)]$$

M: motif structure. C: measured context/activity. Y: fixed-model output.

S(m) is an adjusted score; the Demo reports the same comparison as points lost per removal (member 304: 3.0 vs 0.9).

[2,7] Published sources listed on the final References page.

Adjustment requires measured common causes and comparable support

Common cause

A pre-existing condition affects motif occurrence and the model output. Timing or activity may play that role.

Mediator or consequence

Do not automatically adjust for a variable produced by the motif itself. Measurement timing matters.

Assumptions to check

Causal graph, observed confounding, consistency and overlap. Unmeasured factors or unsupported cells weaken the interpretation.

[2,7] Published sources listed on the final References page.

A teaching example shows how context can inflate a raw difference

Context	Pattern	Count	Mean score
Recent / active	A	80	0.80
Recent / active	B	20	0.75
Older / less active	A	20	0.40
Older / less active	B	80	0.35

Raw: A 0.72 vs B 0.43 (difference 0.29)

Same 50/50 context mix: A 0.60 vs B 0.55 (difference 0.05)

[2,7] Published sources listed on the final References page. Numbers are a teaching example, not experimental data.

The motif library defines which structural hypotheses can be tested

Template

A small labelled interaction graph with an event order. The library supplies candidate hypotheses.

Occurrence

A mapping to actual nodes and edges that satisfies labels, connections and temporal constraints.

Codebooks

CTM records structure; CC records timing; CE records participant properties. A dashed context link is not an extra rating edge.

[2,8] Published sources listed on the final References page.

HNSW retrieves candidates; graph verification establishes valid occurrences

Fast candidate retrieval

Use embeddings and an HNSW index to narrow the search. Nearest-neighbour retrieval is approximate.

Exact validity checks

Verify labels, required graph connections and event order against the original temporal graph.

Remaining limitation

Verification removes invalid candidates. It does not recover valid occurrences that approximate retrieval never returned.

[2,11] Published sources listed on the final References page.

Inverse-probability weights correct sampling within supported populations

Sample probability π(r) = 0.10

Weight w(r) = 10

Sample probability π(r) = 0.50

Weight w(r) = 2

$$w(r) = 1 / \pi(r)$$

Exact retrieval + positive support are required for the unbiased-total result.
Weights do not recover occurrences that retrieval never exposed.

[2] Published sources listed on the final References page.

Faithfulness tests compare input changes with clearly defined controls

Same target

Rerun the same graph component for the same pair preference after removal. Report direction and magnitude.

Control budget

Random, attention and recency edge controls remove three ratings. Report CATGX’s two-rating occurrence separately.

Interpretation

61/63 is an observed paired win count, not population-wide certainty. The 32-search study of a second request (backup) shows the same direction.

Methods: [1,2,9,10]. Experiment: Demo, 63 deletion cases (member 304's request) and 32 (a second request); CATGX and edge-control budgets differ.

The evaluation populations answer different questions

Question	Evaluation population
Where does the ranker fail?	Development: chat 636; regular 848
Does the selected evidence matter?	32 deletion cases; 21 planted probes / 2 patterns
Can diagnosis choose a repair?	343 development diagnoses; 280 held-out failures
Does a repair improve the intended metric?	Compliance: 576 chat requests; quality: 468 requests
Does the cold-start intervention help?	528 new-member requests

All counts describe Demo; the paper benchmarks are separate.

Dataset: MovieLens [5]. Counts: Demo experiment populations, separate from paper benchmarks.

97% measures planted effect removed, not cause-identification accuracy

Known reference

A controlled experiment embeds a specified mechanism, providing a reference contribution to remove.

Reported result

Median fraction of planted effect removed: TemGX 97%, LLM reading history 43%; 21 probes, 2 patterns.

Limit

This checks a controlled mechanism. It does not establish the unique real-world cause of every natural failure.

Method: [1]. Experiment: Demo, 21 probes and 2 planted mechanisms. These percentages are Demo results.

Attention and retrieval require tests suited to their intended use

Attention

A high weight may suggest evidence, but its explanatory meaning depends on the model and the test.

Retrieved context

Similarity or recency helps supply relevant context. It does not by itself demonstrate dependence of a target prediction.

Operational check

Remove selected inputs and compare with controls. The cited attention debate concerns NLP models, not a universal theorem about temporal graphs.

[9,10,1] Published sources listed on the final References page.

Diagnosis retrieval proposes a repair; held-out checks evaluate it

Input and output

Input: a new diagnosed failure. Retrieve similar checked cases and propose a repair using their measured outcomes.

Repair lookup result

343 development diagnoses, 280 held-out failures; working repair selection 37% versus 24% for one global fix.

Separate policy test

“Understand ‘not’ in the request” raises exclusion compliance to 65% versus 46%, with −0.016 NDCG@10. An online test remains separate.

Background: [3,4,5]. Results: Demo offline diagnosis-retrieval and repair experiments.

Semantic-ID and cold-start findings are specific offline starting points

Semantic-ID proxy

Residual k-means codes, not Netflix’s production codebook. Animation/Children codes: 22.9% of wrong-first results vs 13.5% of candidates.

Cold start

Ignoring each title’s latest stranger ratings gives new-member NDCG@10 +0.014; n = 528, CI [+0.010,+0.019].

Next investigation

Explain which historical inputs drive each slice, then test with the team’s representations, logs and success metrics.

Dataset: MovieLens [5]. Results: Demo. Semantic IDs are a residual k-means proxy.

Offline scalability evidence does not define a production latency target

Published stress test

On a separate synthetic graph with one million edges: CATGX 48.2 s; TemGX 254 s, under the reported paper setup.

Practical deployment

Begin with offline failure analysis or sampled regression checks; measure retrieval, encoding and verification costs separately.

Evidence to collect

Recall versus runtime, memory, model-query cost and stability across windows. Choose a serving architecture only after those measurements.

[2,11,3] Published sources listed on the final References page.

My contribution connects scalable search with testable explanations

CATGX

HNSW retrieval, validity checks, stratified sampling and weighted explanation statistics.

TemGX

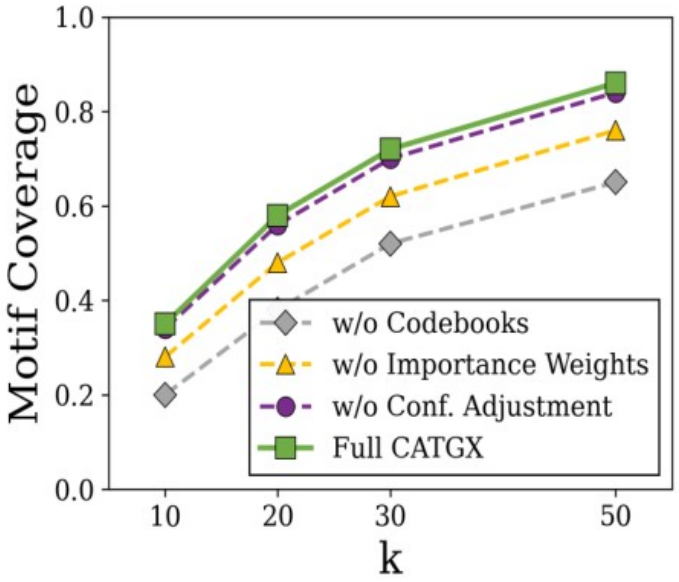
Temporal influence and resistance scoring, compact evidence selection and verification tooling.

Research judgement

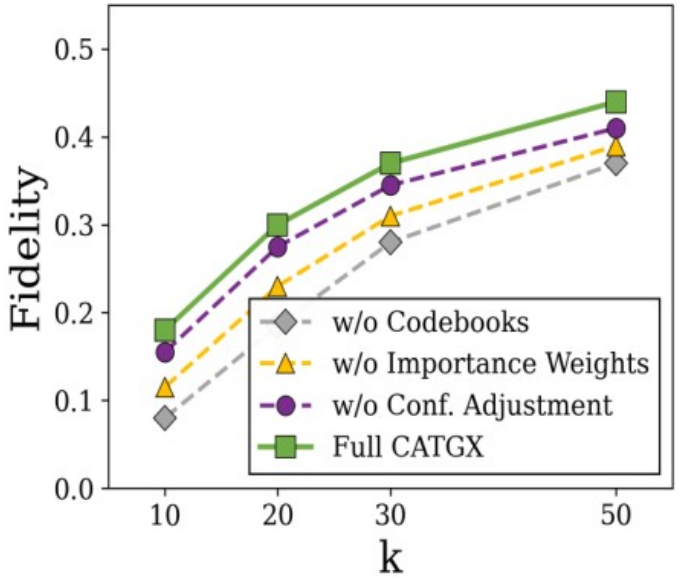
Separate model dependence from behavioural causality. Test failure slices and candidate repairs with explicit assumptions and held-out evidence.

[1,2,12] Published sources listed on the final References page.

Codebooks, weighting and adjustment improve the selected evidence



(e) Ablation k vs Coverage



(f) Ablation k vs Fidelity

1M-edge synthetic graph

Method	Runtime
CATGX	48.2 s
TemGX	254 s
TempME / CoDy / DyGNNExp	OOM

A separate stress test of computational cost

At *k* = 50, UCIM + TGN fidelity: full 0.44, no adjustment 0.41, no weights 0.39, no codebooks 0.37.

[2] Lu et al., KDD 2026, §7.2 and Appendix A.10. 1M edges is a separate synthetic offline runtime experiment.

The evaluation tests explanation quality across three prediction tasks

Task	Datasets and graph meaning	Predictive backbones
Traffic regression	METR-LA: 207 sensors PEMS-BAY: 325 sensors Roads and observed traffic speeds	STGCN, DCRNN
Link prediction	Wiki: user–page edits UCIM: timestamped messages	TGN, TGAT
Event classification	Multihost: audit events Elliptic++: Bitcoin transfers	TGN, TGAT

Controls Matched backbone, temporal neighbourhood, window and explanation budget.

Baselines TGNNExplainer, TempME, CoDy. TGVex for regression.

Measurements Fidelity / AUFSC, explanation size and runtime. Five independent seeds.

[1] Lu et al., ICLR 2026, §5 and Appendices D–F; Table 2.

The motif evaluation separates completeness, faithfulness and cost

Dataset	Interaction	Nodes	Events
Wiki	User–page edit, 172-dim features	9,227	157,474
UCIM	User-to-user message	1,899	59,835

Backbones and controls

TGN and TGAT. 70/15/15 train, validation and test splits. Main comparison: 200 test predictions, $k = 50$.

Baselines

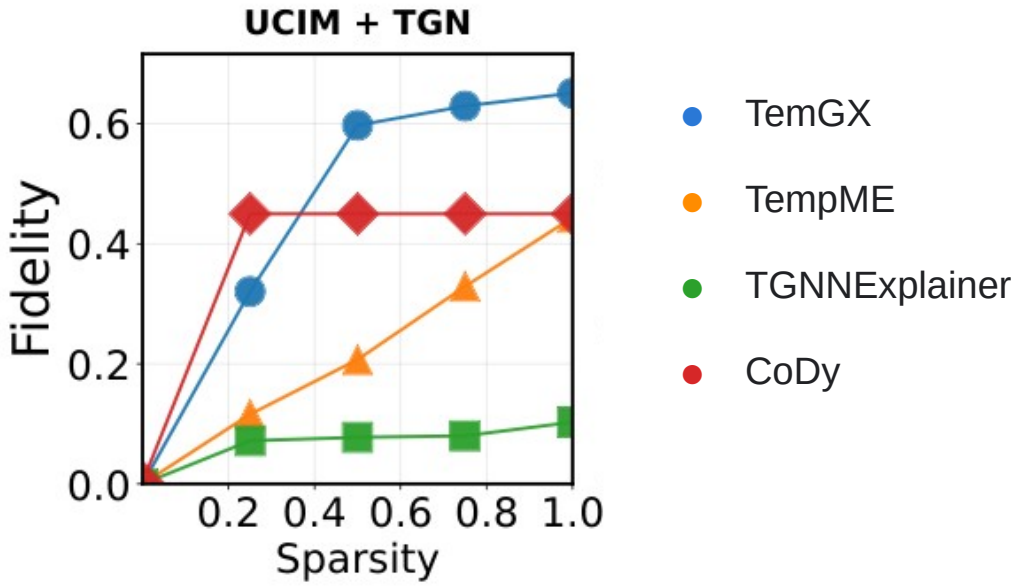
TemGX, TempME, CoDy and DyGNNExp, with aligned backbone, window and size settings.

Coverage counts complete motif occurrences. Fidelity measures model-output behaviour. Runtime is measured separately.

[2] Lu et al., KDD 2026, §7.1 and Appendix A.8.

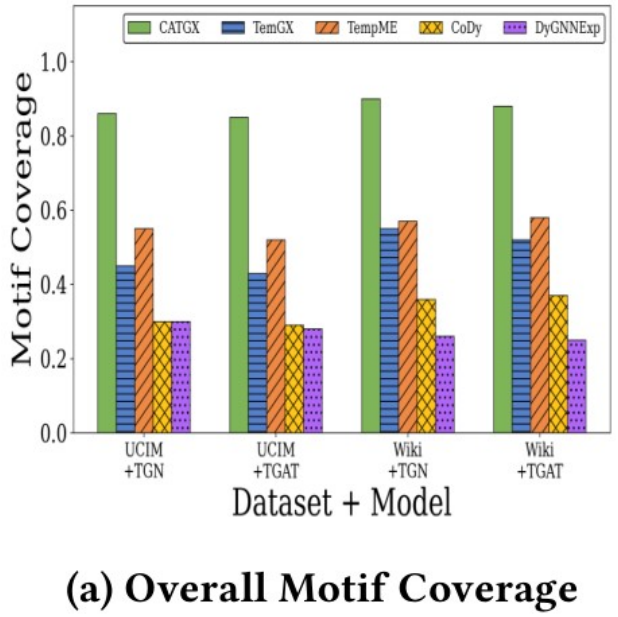
Published results support the explanation methods

TemGX / ICLR 2026



Strong fidelity across budgets

CATGX / KDD 2026

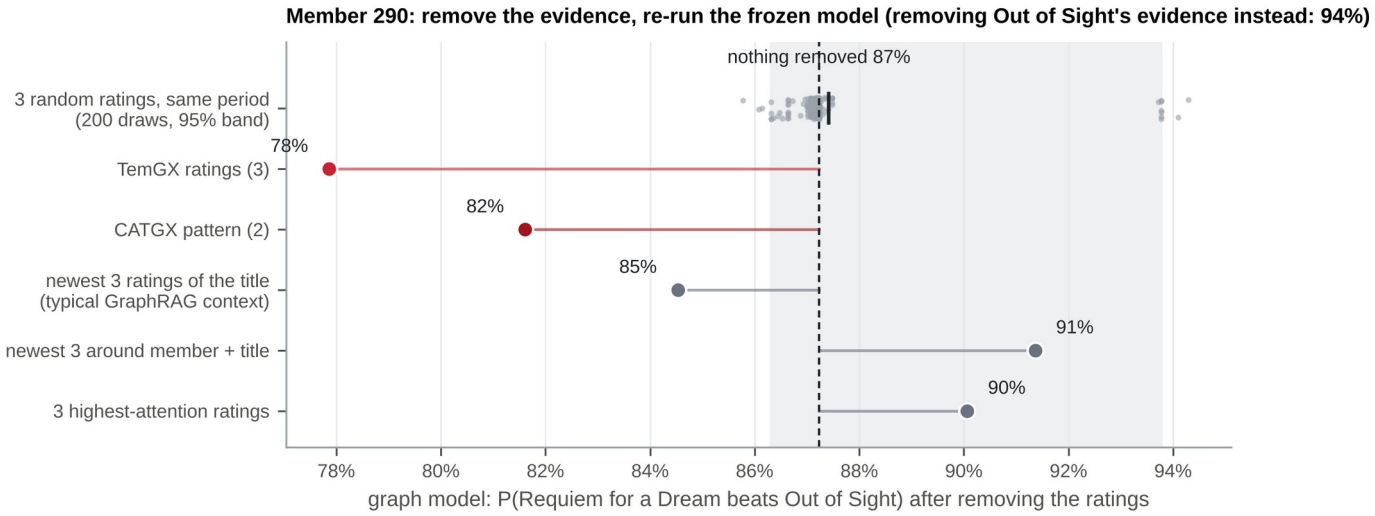


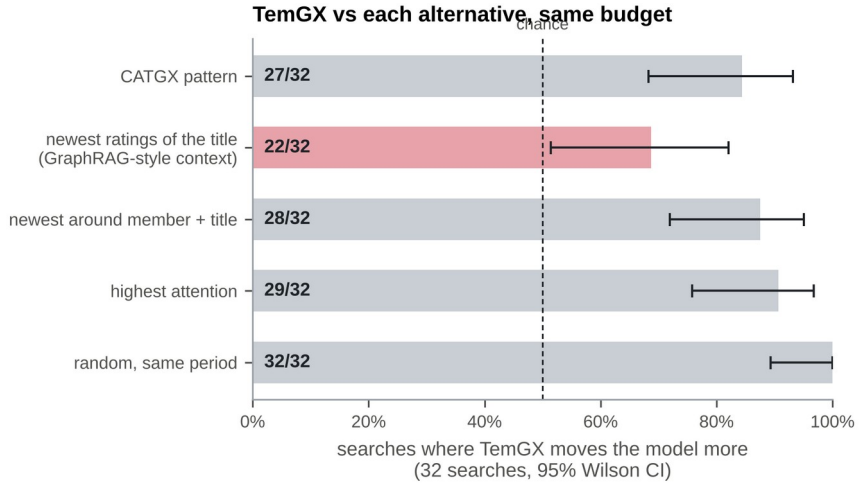
(a) Overall Motif Coverage

High coverage of complete motifs

[1] Lu et al., ICLR 2026, Fig. 4. [2] Lu et al., KDD 2026, Fig. 5(a). Original published panels.

TemGX outperforms the tested deletion controls



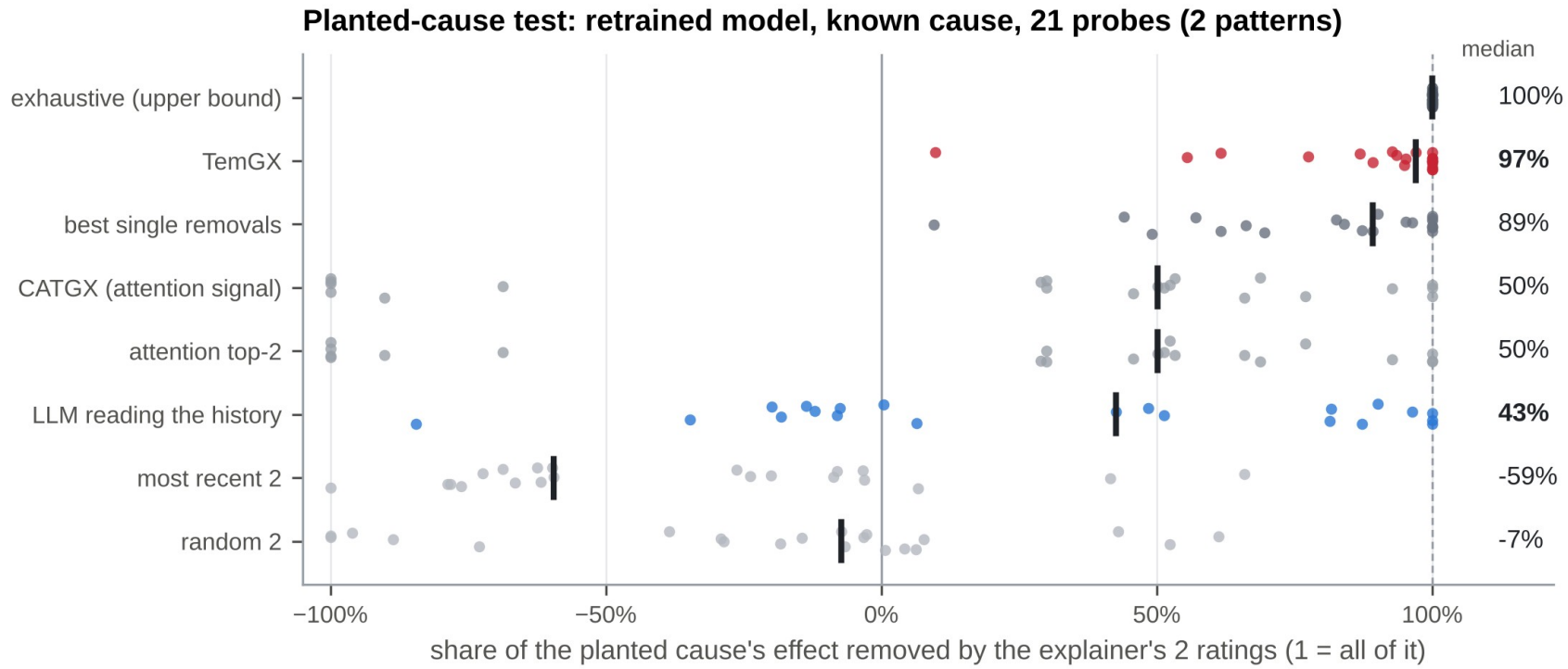


Second case, member 290: random band from 200 draws. Across 32 searches: Wilson 95% intervals. CATGX removes 2 ratings, edge controls 3.

What this gives your team: historical inputs tested against equal-size alternatives.

Method: [1]. Experiment: Demo on MovieLens [5]. CATGX removes 2 ratings; other edge controls remove 3.

TemGX removes 97% of the planted effect

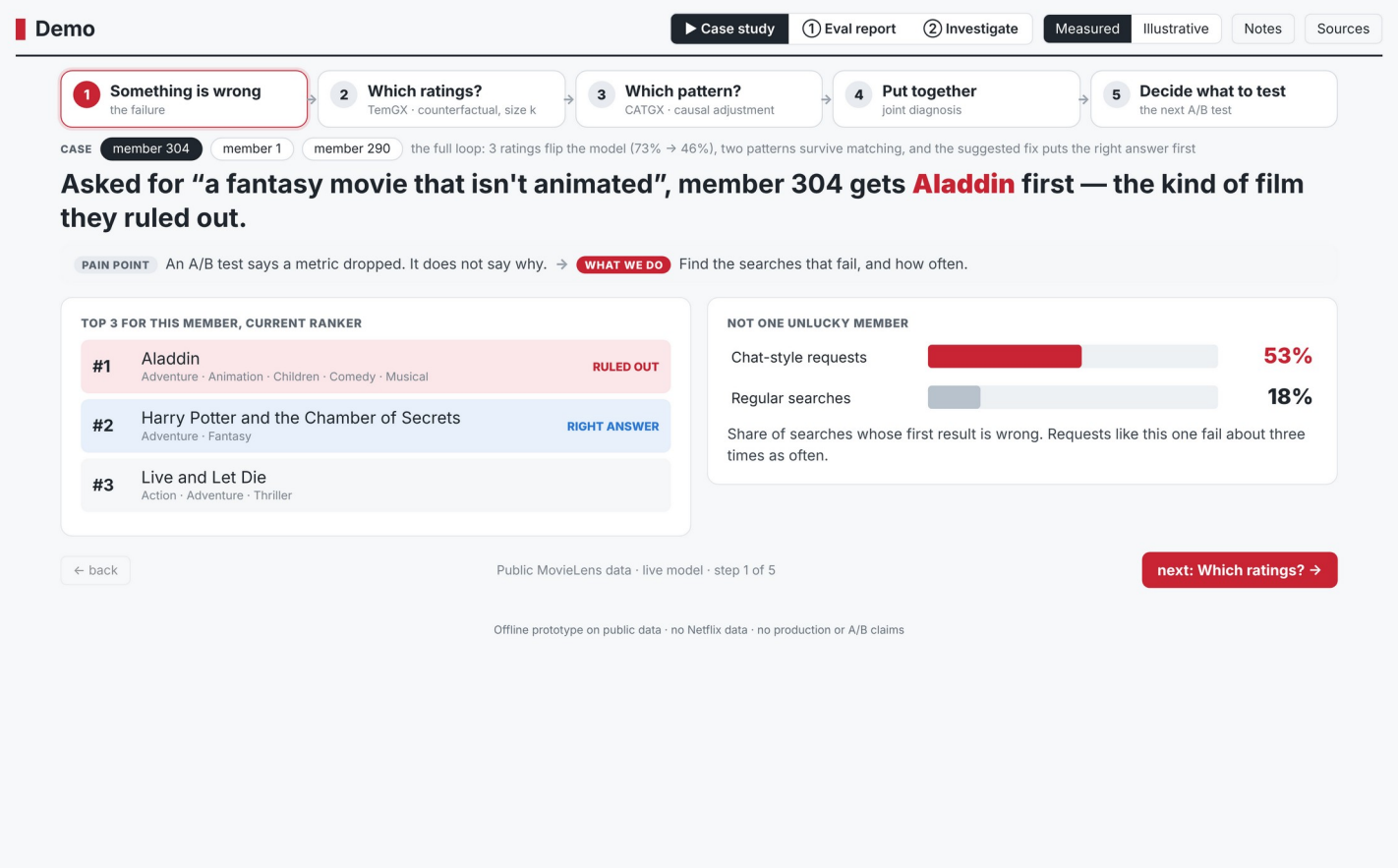


21 probes from 2 planted patterns. Dots = probes. Tick = median fraction of planted effect removed. Exhaustive pairs = upper bound.

What this gives your team: a known reference for testing model-input explanations.

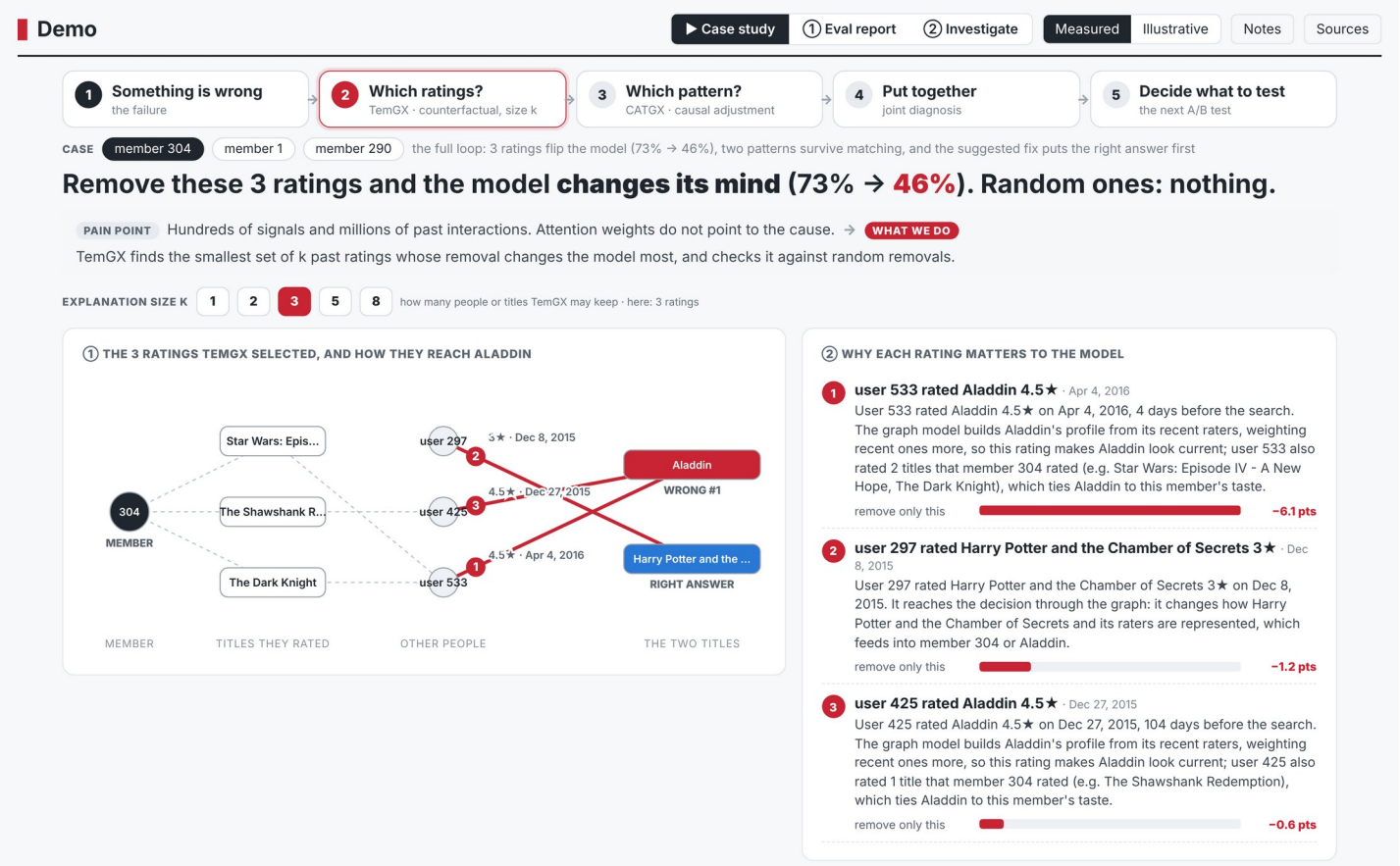
Method: [1]. Experiment: Demo on MovieLens [5]. 21 probes across 2 planted mechanisms; metric is effect-removal fraction.

Demo fallback 1: failed search



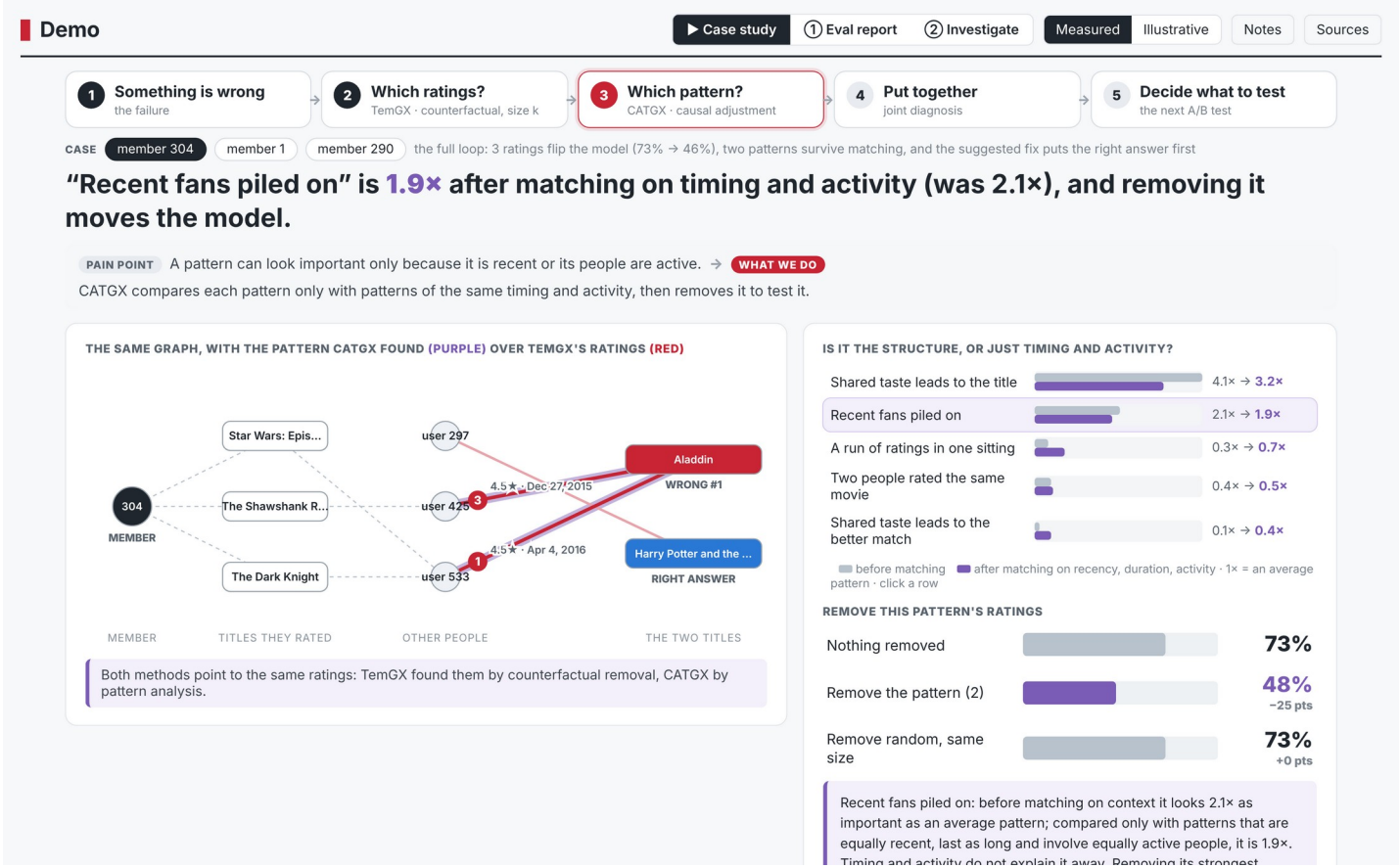
Dataset: MovieLens [5]. Demo screenshot, supplied by the author.

Demo fallback 2: historical evidence



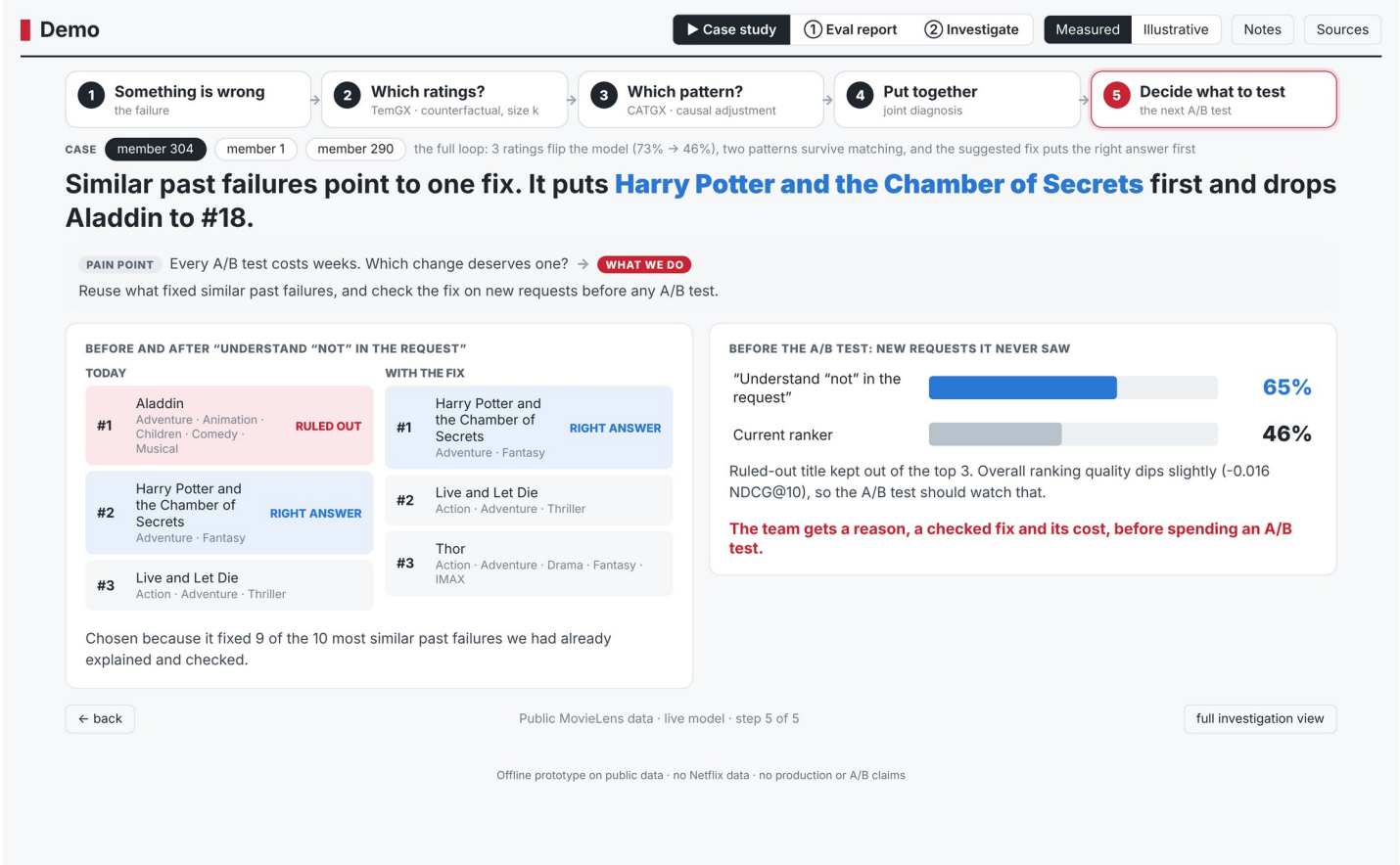
Dataset: MovieLens [5]. Demo screenshot, supplied by the author.

Demo fallback 3: deletion check



Dataset: MovieLens [5]. Demo screenshot, supplied by the author.

Demo fallback 4: repair check



Dataset: MovieLens [5]. Demo screenshot, supplied by the author.

References

[1] Lu, M., et al.

Training-free Counterfactual Explanation for Temporal Graph Model Inference. ICLR 2026.

[2] Lu, M., Vu, H., Le, V. K., Ma, J., & Wu, Y.

CATGX: Causal-Aware Temporal Graph Explanation via Scalable Motif Sampling and Adjustment. KDD 2026.

[3] Gomez-Uribe, C. A., & Hunt, N.

The Netflix Recommender System: Algorithms, Business Value, and Innovation. ACM TMIS 6(4), 2015.

[4] Kohavi, R., Longbotham, R., Sommerfield, D., & Henne, R. M.

Controlled experiments on the web: survey and practical guide. Data Mining and Knowledge Discovery 18, 2009.

[5] Harper, F. M., & Konstan, J. A.

The MovieLens Datasets: History and Context. ACM TiiS 5(4), 2015.

[6] Wachter, S., Mittelstadt, B., & Russell, C.

Counterfactual Explanations without Opening the Black Box: Automated Decisions and the GDPR. Harvard Journal of Law & Technology 31(2), 2018.

[7] Pearl, J.

Causal inference in statistics: An overview. Statistics Surveys 3, 2009.

[8] Paranjape, A., Benson, A. R., & Leskovec, J.

Motifs in Temporal Networks. WSDM 2017.

[9] Jain, S., & Wallace, B. C.

Attention is not Explanation. NAACL-HLT 2019.

[10] Wiegreffe, S., & Pinter, Y.

Attention is not not Explanation. EMNLP-IJCNLP 2019.

[11] Malkov, Y. A., & Yashunin, D. A.

Efficient and Robust Approximate Nearest Neighbor Search Using Hierarchical Navigable Small World Graphs. IEEE TPAMI 42(4), 2020.

[12] Lu, M., et al.
Image Processing Pipeline for Fluoroelastomer Crystallite Detection in Atomic Force Microscopy Images. Integrating Materials and Manufacturing Innovation 12, 2023.

All entries are published articles. Demo results are separate author experiments.

References (continued)

[13] Lu, M., et al.
Context Determines Optimal Architecture in Materials Segmentation. ICLR 2026 Workshop on AI for Accelerated Materials Design (AI4Mat). arXiv:2602.04154.

[14] Lewis, P., et al.
Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS 2020.

[15] Edge, D., et al.
From Local to Global: A Graph RAG Approach to Query-Focused Summarization. arXiv:2404.16130, 2024.